

Research paper

Seasonal forecasting of peak electricity demand using spectral analysis

Eric Augusto Melchor Cruz ^{ID}, Luis Alberto Hernández Armenta, Mohamed Badaoui ^{ID,*}

Instituto Politécnico Nacional, Escuela Superior de Ingeniería Mecánica y Eléctrica, Av. Luis Enrique Erro s/n, 07738, Ciudad de México, Mexico

ARTICLE INFO

Keywords:

Exogenous variable
Demand forecasting
Spectral analysis
Peak electricity demand

ABSTRACT

This paper performs predictions of the Peak Electricity Demand (PED) in the Mexican National Interconnected System (M-SIN) using 400 weekly observations for training purposes. First, this study tackles exploratory data analysis, such as statistical tests, seasonal decomposition, and goodness-of-fit tests. Second, a spectral analysis is carried out to assess cyclical components, stationarity, and white noise presence. Three models are employed to predict 16 future PED values: Seasonal Autoregressive Integrated Moving Average (SARIMA) tool, Prophet, and XGBoost. Alternatively, relying on rigid calendar partitions, each model incorporates an exogenous variable that captures the cyclical position of each observation, enabling improved flexibility and generalization. The Mean Absolute Percentage Error (MAPE), Mean Absolute Scaled Error (MASE), and coefficient of determination (R^2) indicate that the SARIMA model outperforms Prophet, and XGBoost, achieving a MAPE of 1.7333 [%], a MASE of 0.7387 [%], and R^2 of 0.9311. Prophet and XGBoost yield higher errors, with MAPE values of 2.8832 [%] and 2.191 [%], MASE values of 1.2691 [%] and 0.9307 [%], and R^2 scores of 0.846 and 0.8747, respectively. Finally, results from the South of the Mexican Baja California (M-BCS) confirm that the method performs consistently and effectively across different data scenarios, thus supporting its generalization capability and robustness. Specifically, the error metrics and uncertainty intervals from the SARIMA model remain within comparable ranges to those obtained with the former dataset. Therefore, these findings highlight the capability of SARIMA to capture the peak values and to address both short-term and long-term dependencies on the demand trends.

1. Introduction

The use of time series forecasting applied in electrical systems is a practice adopted to assist planning agents in making decisions that bring greater benefit to society, to know the growth in the system demand and the implications of satisfying it at the lowest possible operating cost while maintaining reliability and security, and to consider environmental policies of governmental interests [1]. Moreover, considering that the PED are subject to a non-negligible stochastic nature due to population growth and economic and weather conditions, reliable demand forecasts could enable operators to mitigate risk by ensuring reliability, stability and cost-effectiveness for both the system and clients [2–6]. Therefore, by leveraging the usage of active demand forecasting, the EPS planning strategy could mitigate uncertainties and produce robust and precise decisions.

Atypical growth in the electricity demand prompts a considerable risk exposure on the electric grid and an increasing requirement for grid resilience, especially during heat waves [7]. The situation above and an

upward trend, as seen in recent years, raise the question of to what extent the Electrical Power System (EPS) would be able to preserve its reliability without having blackouts [2]. The electricity grid is sensitive to fluctuations, and instability can result in outages where the social impact is substantial. The aforementioned characteristic implies that EPS must ensure supply and demand are balanced at any given time, requiring additional generation capacity to protect the system against supply inadequacy. The additional generation is the reserve requirement, generally a percentage of the demand. Consequently, the future probable demand directly impacts the operation planning procedures as shown in [1] and [8].

The literature about electricity demand strongly suggests that long-term forecasting is at the outset of tackling the planning, operation, and developing potential investments in generation, transmission, and distribution equipment [9–11]. Furthermore, strengthening the national energy and electricity demand modeling implies that future grid expansion and operation decisions are reliable. Finally, according to [5], three directions suggest an improvement in demand modeling:

* Corresponding author.

E-mail address: mbadaoui@ipn.mx (M. Badaoui).

1. Annually electricity demand forecasting by sector and then an hourly disaggregation.
2. Bottom-up approach with either stochastic or average load profiles.
3. Soft-linking long-term energy system modeling tools considering short-term power market models

Considering that atypical climate conditions are one of the underlying factors influencing the PED and that Bottom-up approaches oversimplify the impacts of future climate changes on meteorological conditions, the performance of the Bottom-up method is diminished by this drawback. On the other hand, due to the high computational cost and lack of reliable, long-term meteorological data, load modeling is often reliant only upon self-contained time series data [7,12,13]. Therefore, the present paper tackles the first part of annual electricity demand forecasting.

However, when comparing our results to those of older studies such as [2] and [12], it must be pointed out that we intend to bolster our model using a normalized cycle position index to represent the position of the observation over a year horizon instead of an intrayear seasonality or similar months approach, respectively. Unlike [2], the results in the present paper are a forecast with weekly discretization rather than daily. Those above aim to circumvent substantial amplification error and improve the model's accuracy [14]. Finally, given the need for more reliable weather information, the present paper mainly employs historical data.

The proposed methodology in [3] has been used to forecast the probability distribution of annual and weekly peak electricity demand. Moreover, the implemented demand model is split into annual effects and half-hourly effects, with the two parts of the model estimated separately. Nevertheless, the authors assume a stable long-term temperature pattern. Consequently, given that it closely follows the paradigm of time series forecasting, the PED prediction attains the consistent behavior of weather conditions.

Authors in [6] introduce a novel closed-loop neural network framework for next-day and peak load forecasting. The study employs a cascade ensemble approach, combining wavelet transform (WT), mutual information-interaction gain (MI-IG), and neural networks. The wavelet transform decomposes the load data into distinct frequency components to analyze underlying patterns. Regarding feature selection, MI-IG ensures informative and relevant variables are included in the forecasting model. The framework was validated on two real-world datasets: the PJM electricity market and the Austrian national load database (2015–2020). Furthermore, the reported errors in this study are a Mean Absolute Error of 224.17 [MW], a Root Mean Square Error of 37.87 [MW], and a Mean Absolute Percentage Error of 0.29 [%] for the PJM dataset on June 10, where the peak value is 93 [MW]. In contrast, the Austrian dataset on September 30 has a Mean Absolute Error of 40.56 [MW], a Root Mean Square Error of 40.56 [MW], and a Mean Absolute Percentage Error of 0.55 [%], where the maximum electrical load is around 8000 [MW].

AI techniques disrupt the domain of time series forecasting. In [15] the problem of accurate energy forecasting by an intelligent hybrid model is proposed. The model integrates feature selection, signal decomposition, and deep learning techniques. The three key components are based on mutual information-based feature selection, a variational mode decomposition (VMD) approach to decompose the original energy consumption signal into intrinsic mode functions (IMFs), and a Long Short-Term Memory (LSTM) neural network to perform time-series forecasting. Moreover, [16] reports an enhanced forecasting model to manage volatility using an optimized XGBoost algorithm with integrated error correction features. The proposed model achieves a symmetric Mean Absolute Percentage Error ranging from 1.77% to 11.41% throughout the study period. The authors compare the enhanced XGBoost with standard XGBoost and ARIMA + LSTM. On the other hand, hybrid models integrating ARIMA have been reported in the literature, such as [17], in which authors explore Prophet, the Vector Autoregressive Integrated

Moving Average (VARIMA), and a combination of ARIMA with the Prophet model.

Various machine learning methods are often utilized due to their ability to predict. For example, in [14], global horizontal irradiance (GHI) is predicted using Machine Learning (ML) regressor algorithms such as Multiple Linear Regressor (MLR), Decision Tree (DT), Random Forest (RF), Gradient Boost (GB), Light Gradient Boost Machine (LGBM), and Extra Tree (ET). Furthermore, this study identifies a subset of the relevant input variables for the prediction of GHI by applying two methods, Feature Combination (FC) and Feature Selection (FS). Conversely, due to the multi-collinearity phenomenon, we try to circumvent redundant predictor variables so as not to decrease the model's precision [14]. This selection process is analyzed rigorously in [18], where feature selection techniques such as filter, wrapper, and embedded are computed and compared to reduce the dimensionality of the input features in the models.

In line with previous studies, in [19–21], a SARIMA model is exhibited to assess the future electricity demand. Authors in [19] present a model with a seasonality parameter of 144, representing the 10-minute intervals of a day. Finally, the forecast results in the demand values on a two-day ahead horizon. Contrary to [19–21], in the present paper, the proposed model employs an exogenous variable and weekly historical data. Furthermore, it is worth noticing that in [21], the dataset is based on the power consumption of one apartment, and it does not reflect the complexity of forecasting demand of a large sector with penetration of uncertainty. On the other hand, auto-SARIMA has been proven to outperform models such as Prophet and Neural Prophet regarding long term estimation of global horizontal irradiance [22].

Conventional seasonal models such as HOLT-WINTERS and standard SARIMA underperform when facing irregular cycles and complex nonlinearities. These methods rely on fixed seasonal patterns and lack adaptation to evolving trends and anomalous behaviors, restricting their predictive robustness under real-world variability. For example, regarding the literature in time series forecasting, the MAPE errors reported in [10] for the SARIMA, HOLT-WINTERS, LSTM, and Autoregressive Integrated Moving Average (ARIMA) models are 6.05%, 9.18%, 10.67%, and 13.87%, respectively. The data for training the SARIMA model in [10] ranges from 2008 to 2018, and 2 years are predicted. Authors in [12] report a minimum MAPE of 1.65% for the January forecast experiment and 2.21% MAPE for October, where similar months-based SARIMA are employed to predict electricity demand in the short term. The methodology used in [12] is based on a rolling six-hour ahead forecasting, i.e., once the observations are available, the model training set is updated, and the model is fitted anew. Results in [23] where various features such as temperature, humidity, cloud cover, visibility, wind speed, UV index, and dew point are incorporated, SARIMA and LSTM models obtain MASE errors of 0.5824% and 0.32456%, and MAPE errors of 5.2369% and 3.22134%, respectively. Nevertheless, including the aforementioned exogenous variables could lead to a computational burden during the preprocessing process. Moreover, the reliability of the data could be compromised due to the inherent uncertainty within the feature itself.

In Mexico, forecasting methods are maturing the research topic with a wealth of well-understood methods and algorithms applied to planning and designing applications. Such approaches are based on artificial intelligence, statistical, and stochastic models as presented in [24] and [25]. The former is applied to the region of M-BCS, a sector with a small demand compared to M-SIN. The research employs two years of hourly data to predict the following 14 years. Furthermore, the authors report that the more neurons, the better the forecast. Nevertheless, the ANN employed could be overfitted, raising concern about the extent to which the model could perform accurately. The latter presents a discrete and continuous approach. The continuous model is based on the Ornstein-Uhlenbeck stochastic process with a deterministic function, whereas the discrete model employs an Integrated Autoregressive Double Seasonal Moving Average process. The authors only predict a single day under a

set of possible future scenarios. On the other hand, the methodology described in [26] is a bottom-up and top-down technique to forecast the demand for long-term time horizon. Finally, in Mexico, it exists a record of proficient application of stochastic differential equations and memoryless transformation, albeit applied in wind speed, contributes to the analysis in time series as shown in [27].

The literature on applying forecasting models has raised the validity of time series forecasting using either statistical or artificial intelligence-based algorithms. Consequently, it is crucial to ensure a comprehensive approach by balancing quantifiable metrics and qualitative factors in decision-making and assessing the solution's feasibility based on the demand's characteristic behavior, regardless of the algorithm to implement.

1.1. Contributions

Demand as a time series exhibits a nonlinear and dynamic component, reflecting fluctuations and abrupt changes in consumption magnitude. Moreover, the demand levels as a signal contain features such as slopes, periodic patterns, and anomalies, which challenge accurate modeling [6]. Detecting sudden variations in time series data is a key challenge in many forecasting applications, including energy systems. In the context of electricity demand in Mexico, two types of abrupt changes are particularly relevant: outliers and peaks. While both represent deviations in the data, they serve different purposes in analysis. Outliers refer to unexpected demand values that significantly deviate from the typical range, possibly due to anomalies. On the other hand, peaks are expected but critical demand events, such as the highest electricity consumption periods in a week or season. These values are important for grid planning and operational decisions, even if they follow a regular pattern. A peak typically corresponds to a local or global maximum within the demand time series. However, it may not always appear as a sharp spike or an outlier, especially in Mexico, where electricity demand may exhibit repetitive, seasonal, consistent peaks that are still crucial for forecasting. This makes peak detection a nuanced task. Analysts must ask: How much higher must a demand value qualify as a peak? Moreover, how do we define a local maximum amid natural fluctuations? This is further complicated by noise in real-world data, arising from minor measurement inconsistencies, weather-related variability, or consumer behavior. This is especially important in the Mexican electricity demand setting, where climatic, social, and industrial patterns influence recurring high-demand periods, making accurate forecasting of these peaks essential for avoiding grid stress or generation shortages.

When dealing with real-world data, the gap between the model and reality enhances the need to identify a forecast model that balances quantifiable metrics with qualitative parameters and exogenous factors. This work contributes to the PED forecasting literature by applying and comparing a set of methods tailored to the unique dynamics of the M-SIN, introducing a robust evaluation pipeline for model validation, and providing a generalizable forecasting framework that can inform operational deployment in various data contexts such as the M-BCS region. Furthermore, this study's notable contribution is its thorough focus on taking data, concepts, and modeling of time series and making them observable, which raises concern about the implications when dealing with sensitive results, such as the Peak Electricity Demand.

The present research proposal is a weekly discretized approach under an annual stationarity condition and an immersion of a single exogenous variable to preserve the principle of parsimony and not to overfit the model. Including a normalized cycle position index as an exogenous variable is a parsimonious yet effective proxy for capturing annual seasonality in electricity demand. This approach is motivated by incorporating periodic information without relying on external climatic or calendar-based variables, which may be unavailable or unreliable, or introduce additional modeling complexity. Unlike categorical variables such as month or week-of-year, the normalized index offers a continuous and smooth representation of the observation's position within the

annual cycle. This enables statistical and machine learning models to more naturally learn cyclical patterns in the data, particularly in systems where demand exhibits strong seasonality but is not strictly aligned with calendar months or holidays as is the case for the M-SIN. Moreover, this normalized index avoids overfitting by not introducing a large number of categorical dummy variables, and it retains interpretability, making it especially suitable for models where transparency and computational efficiency are valued, such as SARIMA or regularized regressors like XGBoost. This variable is further supported by the spectral analysis, which reveals a dominant frequency at approximately 52 weeks, justifying the relevance of including a seasonality-related index. Within the framework of these criteria, the main contributions of this research can be outlined as follows:

- The proposed methodology in this study emphasizes the use of real-world data, minimal interaction with exogenous inputs, and interpretable statistical modeling to provide a practical, accurate forecast. The intention of implementing SARIMA, Prophet, and XGBoost is to demonstrate that, even if well-established literature of time series forecasting exists, it is only through its realization into real-world data that its impact can be assessed and that theories can be refined based on the empirical evidence. Therefore, implementing the aforementioned forecasting models supported by spectral analysis contributes to a growing body of research showing that working with real-world data leads to extensive studies so that one can achieve some degree of certainty in the results the model is producing.
- The analysis exhibited in this paper underscores the importance of selecting an appropriate discrete seasonal model that aligns with a probable future trajectory and planning utilities. Furthermore, this study's contribution is its thorough focus on the hyper-parameters and exogenous variable selection to improve the model's adaptability. The results of the proposed approach show that the SARIMA model can lead to more robust and practical decisions, enhancing them to anticipate and mitigate likely potential issues in the future and make proactive decisions.
- To our knowledge, no study has yielded a spectral analysis as the one presented in the present paper, where raw data and results from the model's fitting have been assessed in such depth. To rectify the decision-making regarding the hyper-parameters, we performed a study based on the Fourier transform and the periodogram. It must be pointed out that there is no direct interaction between the models and the spectral analysis. Both procedures are done independently. The spectral analysis, specifically the Fourier transform and periodogram, revealed a dominant frequency corresponding to a 52-week cycle. This finding directly informed the selection of the seasonal period parameter $s = 52$ in the SARIMA model, ensuring the model aligns with the observed dominant periodic behavior in the data. Moreover, the strength and clarity of the annual peak in the spectrum supported the decision to include seasonal autoregressive and moving average terms in the model structure. By identifying this strong seasonality in the frequency domain before fitting the model, we reduced the risk of under- or over-specifying the seasonal components. This spectral insight helped constrain the search space for the seasonal hyperparameters P, D, Q , thereby improving both modeling efficiency and robustness.

The present paper is organized as follows, Section 2 gives an overview of the exploratory data analysis performed in the time series under study. Section 3, 4 and 5 provide identification techniques for the seasonal time series model and an analysis of the model performance of in sample predictions from SARIMA, Prophet and XGBoost model, respectively. Finally, Section 6 and 7 reports the analysis of the results from out of sample forecast and conclusions, respectively.

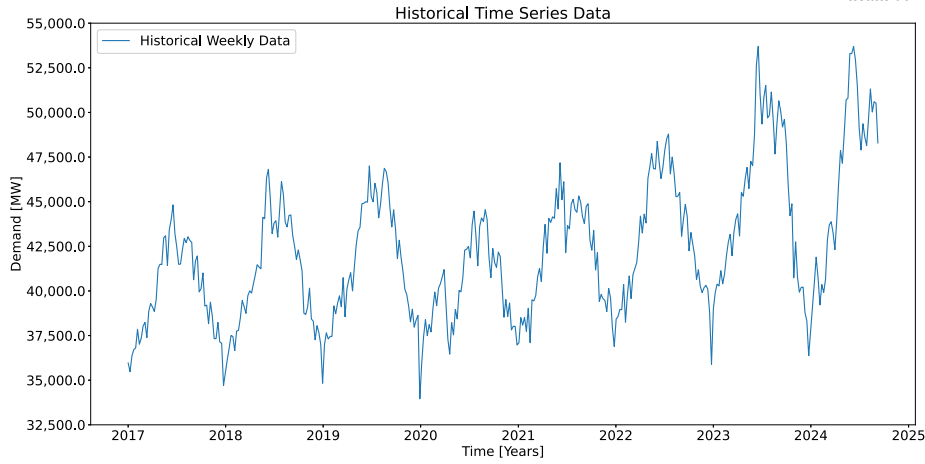


Fig. 1. Raw Data.

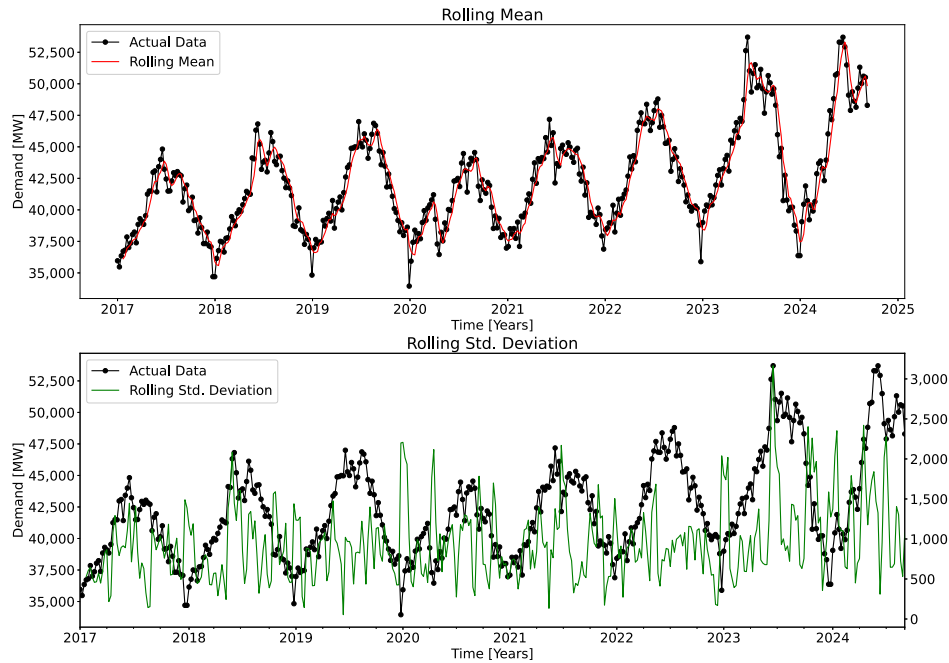


Fig. 2. Rolling Mean and Std. Deviation.

2. Exploratory data analysis

Exploiting the unique patterns in historical data, such as seasonality, trends, and hypothesized exogenous variables, is crucial for electricity demand forecasting. The following experiments are based on the time series of the PED in the Mexican National Interconnected System [28], where the data recorded comprehends from January, 2017 to the first week of September, 2024, as depicted in Fig. 1.

Fig. 1 shows that, even in presence of a weekly and annual periodic behavior similar between observations, there are considerable variations in magnitude that might be a result of the combination of multiple random factors. The rolling statistics of the M-SIN's PED are shown in Fig. 2 where the rolling mean and standard deviation (std. deviation) consider a window of 4 data points, a general constitution of weeks in a month. We perform the following graphical analysis to visualize patterns and variability.

Results from Fig. 2 show that the rolling mean smoothens out the fluctuations in the actual data and defines a rising trend, validating the result depicted in Fig. 4 for the trend subplot. Conversely, the peaks in

the rolling std. deviation describes the volatility or variability of the time series. Mainly, the M-SIN's PED presents peaks in the rolling statistic over periods with a peak in the percentage by drought, as seen in Fig. 3, or a new year begins.

Fig. 4 reveals further insights into the time series of the PED, the assessment of these results is to separate distinct patterns in the data for more in-depth understanding and modeling.

We decline the idea that the outliers lower than the average values observed in the Box plot from Fig. 4 result from major failure events in the system, given that the PED is greater than the one under the failure condition. It is worth mentioning that neither PED forecast considers events where atypical failure events occur, nor is it introduced in the modeling. On the other hand, the Box plot from Fig. 4 depicts that the demand has an augmentation in the middle months, plausible variability, and high interquartile ranges, whereas, toward the end of the year, the demand tends to lessen, acquiring stability. Finally, the PED increase in the middle weeks of the year is a consequence of the intense heat waves that took place in Mexico during that range of dates, and

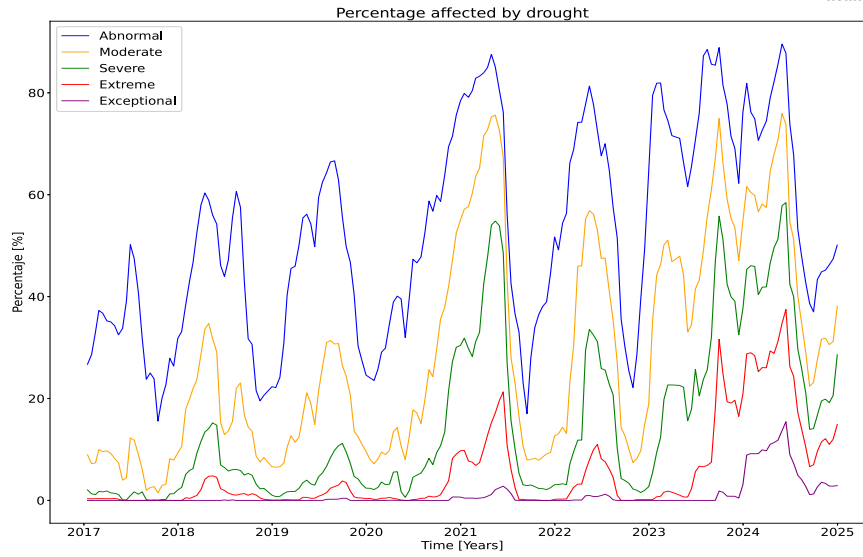


Fig. 3. Percentage of drought impact.

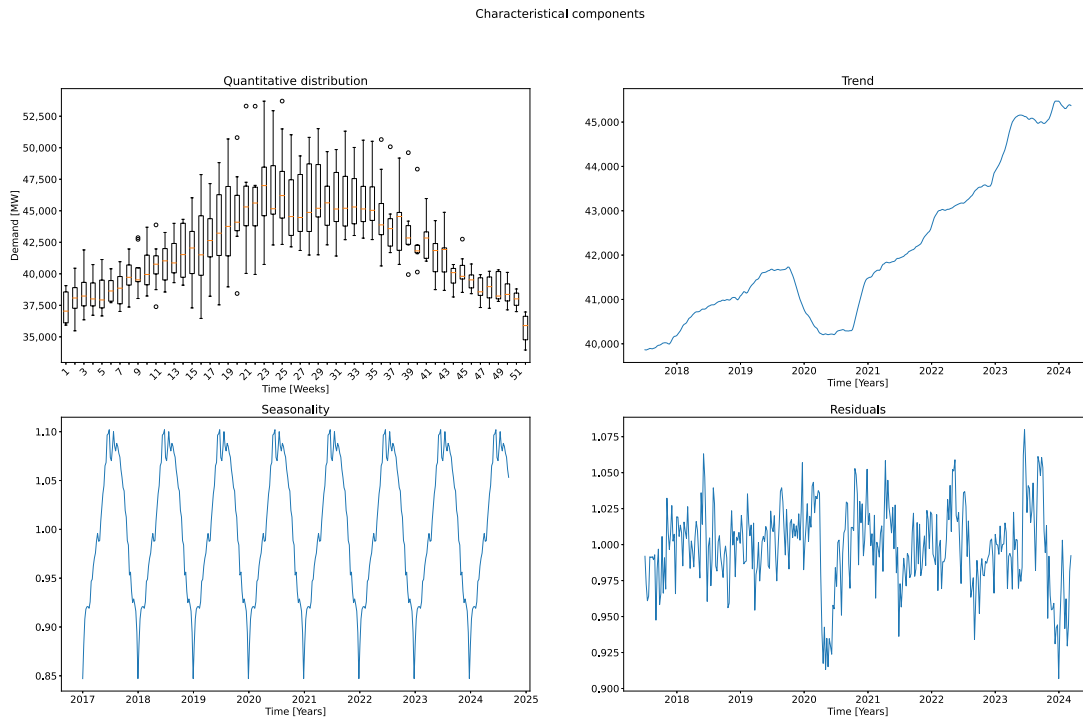


Fig. 4. Quantitative descriptions and seasonal decomposition.

it can be demonstrated by the exceptional drought displayed in Fig. 3 [29].

On the other hand, from the seasonal decomposition, one can observe a steady increase in the trend, with a substantial decrease around 2020 due to effects of COVID-19. The seasonality component describes clear periodic fluctuations with a cycle that repeats annually indicating a strong seasonal pattern in the series, i.e., we can infer the seasonality s of 52. The residuals show a consistent fluctuation around zero with no evident pattern. Consequently we conclude that the decomposition provides a useful insight of the main components of the series. Nonetheless, spikes in the residuals in 2023–2024 demonstrate the presence of irregular factors affecting the demand and not being captured by the trend or seasonality models.

3. SARIMA model

3.1. Preliminaries

The steps in model identification and estimation can be summarized as follows:

Preliminary Transformations. For this procedure to be valid, the data must represent a realization of an ARMA process and, more specifically, a stationary process. If the data exhibits nonstationary features such as trends or seasonality, transformations are required to ensure stationarity [30]. Nonstationary behavior can be identified through visual inspection

Table 1
Unit Root Tests.

Test	Null hypothesis (H_0)	Alternative hypothesis (H_1)	Criteria to reject H_0
ADF	Series has a unit root	Series has no unit root	Test statistic < Critical Value; p-value < 0.05
KPSS			Test statistic > Critical Value; p-value > 0.05

of the time series plot. Trends are typically indicated by a slowly decaying autocorrelation function, while seasonality is revealed by an autocorrelation function with nearly periodic patterns. To address these issues, two common methods can be employed: seasonal decomposition, which separates the series into trend, seasonal, and residual components, or differencing, which removes these nonstationary elements directly [30].

Identification and Estimation. Let $\{X_t\}$ be the transformed series. The problem is to find the most satisfactory to represent $\{X_t\}$. Albeit, one might infer that by choosing higher values for p and q , the resulting fitting model improves its performance. However, estimating numerous parameters introduces estimation errors affecting adversely the model's prediction [30]. Therefore, the p and q value election is suggested to be based primarily on the minimization of either Akaike's Information Criterion (AIC) [10] computed by:

$$\text{AIC} = 2(p + q + 1) - 2 \ln L(\phi, \theta, S(\phi, \theta)/n) \quad (1)$$

where $L(\phi, \theta, \sigma^2)$ is the likelihood of the data under the Gaussian ARMA model with parameters (ϕ, θ, σ^2) and Mean Square Errors $r_0 \dots r_{n-1}$ defined by:

$$L(\phi, \theta, \sigma^2) = \frac{1}{\sqrt{(2\pi\sigma^2)r_0 \dots r_{n-1}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{j=1}^n \frac{(X_j - \hat{X}_j)^2}{r_{j-1}} \right\} \quad (2)$$

and $S(\phi, \theta)$ is the residual sum of squares defined as:

$$\sum_{j=1}^n (X_j - \hat{X}_j)^2 / r_{j-1} \quad (3)$$

where $\hat{X}_j$ is the one-step predictor of X_j .

Once a model has been found that minimizes the AIC value, an assessment under a goodness of fit assumption is required. The aforementioned is by checking that the residuals are like white noise. For any values of p and q , the maximum likelihood estimates of ϕ and θ are the values that minimize the AIC. The smallest value of AIC indicates the combination of p and q values to define the model [30]. Moreover, graphical inspection methods are convenient to identify the possible values of p and q . Hence, the provisional order selection of a model can be drawn from the inspection of the sample Autocorrelation Function (ACF) of $\{X_t\}$ from:

$$\hat{\rho}_X(h) = \frac{\hat{\gamma}_X(h)}{\hat{\gamma}_X(0)}, \quad (4)$$

where $\hat{\gamma}_X(h)$ is the sample autocovariance function. On the other hand, the Partial Autocorrelation Function (PACF) of the stationary series for any set of observation $\{x_1, \dots, x_n\}$ with $x_i \neq x_j$ for some i and j , the PACF $\hat{\alpha}(h)$ is defined as follows:

$$\hat{\alpha}(0) = 1 \quad (5)$$

and

$$\hat{\alpha}(h) = \hat{\phi}_h, \quad h \geq 1 \quad (6)$$

where $\hat{\phi}_{hh}$ is the last component of

$$\hat{\phi}_h = \hat{\Gamma}_h^{-1} \hat{\gamma}_h \quad (7)$$

$\hat{\Gamma}_h = [\hat{\gamma}(i-j)]_{i,j=1}^h$ and $\hat{\gamma}_h = [\hat{\gamma}(1), \dots, \hat{\gamma}(h)]^T$. Both functions are estimated for each lag h [25][30].

On the other hand, SARIMA models identify the values of d and D for a possible transformed data so as to make the differenced observations stationary in appearance.

$$Y_t = (1 - B)^d (1 - B)^D X_t \quad (8)$$

Consequently, there exists an examination of the ACF and PACF of $\{Y_t\}$ at lags that are multipliers of s for an indication of order P and Q in the model [30].

3.2. Hyper-parameter identification

Let $\{X_t\}$ be the time series and s be seasonal component period. Consequently, differentiating the $\{X_t\}$ at lag s , the seasonal component of period s is eliminated. When fitting an ARMA(p, q) model:

$$\phi(B)Y_t = \theta(B)Z_t$$

to the differenced series $Y_t = (1 - B^s)X_t$, then the model for the original series is $\phi(B)(1 - B^s)X_t = \theta(B)Z_t$ [30].

It is worth noticing that the process Y_t is casual if and only if (iff) $\phi(z) \neq 0$ and $\Phi(z) \neq 0$ for $|z| \leq 1$. In applications, it is rarely employed values of $D > 1$, $P > 3$ and $Q > 3$ [30]. The casual ARMA process, satisfied by the differenced process Y_t , is equivalent to:

$$\phi^*(B)Y_t = \theta^*(B)Z_t \quad (9)$$

where $\phi^*(\cdot), \theta^*(\cdot)$ are polynomials of degree $p + sP$ and $q + sQ$, respectively, whose coefficients could be expressed in terms of $\phi_1, \dots, \phi_p, \Phi_1, \dots, \Phi_p, \theta_1, \dots, \theta_q$, and $\Theta_1, \dots, \Theta_Q$ [30].

The presence of stationarity in the time series arises the premise to employ a univariate test for the presence of a Unit Root, and thus to determine non-stationarity in the time series. To be fluent in when performing time series analysis, the Augmented Dickey-Fuller (ADF) test along with Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test is employed to determine the value of d and D . The null hypothesis for each test is as shown in Table 1.

The stated identification techniques altogether with the proposed spectral analysis to support the decision-making are stated in Fig. 5.

3.3. Results

The interpretation of p-value in the KPSS test when being lower than the significance level (0.05) concludes that the series is non-stationary. Whereas in ADF test, it would mean the tested series is stationary. Therefore, Table 2 and 3 describe the statistics of the series under an assumption that the data either has no differentiation or is differentiated once.

From the results in Table 2 and 3, we observe that for the case when there no exists differentiation the ADF test concludes stationary, contrary to KPSS, exhibiting a non-stationarity around the mean. From this standpoint results yields to consider that the series is difference stationary. Regarding this assumption one can observe that both test conclude

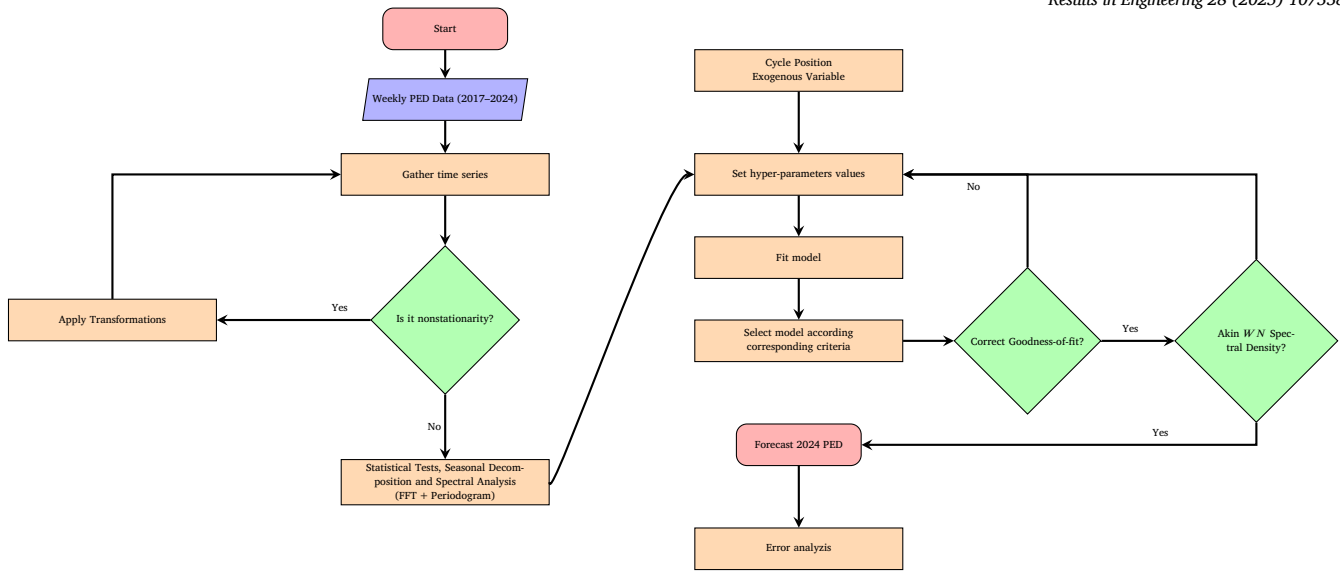


Fig. 5. Examination Process.

Table 2
Augmented Dickey–Fuller Test.

Param	Values	
	No Differentiation	One Differentiation
ADF Statistic	-4.447671	-5.192675
p-value	0.00024	9.0999e-06
Critical Value at 1%	-3.447	-3.447
Critical Value at 5%	-2.869	-2.869
Critical Value at 10%	-2.571	-2.571
Statistical Property	Stationary	Stationary

Table 3
Kwiatkowski–Phillips–Schmidt–Shin test.

Param	Values	
	No Differentiation	One Differentiation
KPSS Statistic	0.836311	0.020361
p-value	0.01	0.1
Critical Value at 1%	0.347	0.347
Critical Value at 2.5%	0.463	0.463
Critical Value at 5%	0.574	0.574
Critical Value at 10%	0.739	0.739
Statistical Property	Non-stationary	Stationary

that stationarity under a single series' differentiation. Therefore, the value of both d and D is set to 1.

The p , q and P , Q parameters selection are drawn from the inspection of the sample ACF and PACF of the stationary series, Fig. 6, i.e., the series under one differentiation.

From Fig. 6, it is depicted that a significant positive spikes in ACF and PACF plot cut at lag 4, underpinning the application of an ARIMA(p , d , q) process. Furthermore, as suggested in [30], as one positive correlation coefficient in the sample PACF is statistically significant, we opt to set p as 1 as first iteration for the modeling, further evaluation of p taking values up to 3 is carried out. Unlike the values of p , q is defined by the sample ACF and considers a single correlation coefficient outside the confidence interval. Similarly, given the 52-seasonal parameter, we make use of the first seasonal offset observation in the model, then $P = 1$. Conversely, when observing the sample ACF, no significant correlation coefficient is observed, and consequently, Q is defined as 0.

Regarding stationary processes, the spectral representation of $\{X_t\}$ provides a frequency-domain perspective analogous to the Fourier representation of deterministic functions. Spectral analysis, also known as

frequency domain analysis, offers an alternative to time domain analysis. While both approaches are mathematically equivalent, the frequency domain often reveals insights that are less apparent in the time domain [30,31].

The idea of representing a time function as a combination of sinusoidal components is highly applicable to random processes. Unlike deterministic functions, a sample function of a random process is drawn from an ensemble of possible time functions. Consequently, the weighting function, or *spectrum*, must describe the average rate of change across this ensemble. Spectrum estimation of discretely sampled processes is usually based on procedures employing the Fast Fourier Transform (FFT) where the periodogram of the observations $\{x_1, \dots, x_n\}$ is introduced as a function given by [30–33]:

$$I_n(\lambda) = \frac{1}{n} \left| \sum_{t=1}^n x_t e^{-it\lambda} \right|^2 \quad (10)$$

Spectral analysis represents a time series $\{X_t\}$ as a sum of sinusoidal components with uncorrelated random coefficients, providing a decomposition that reveals the frequency-based structure of the series [30–32]. Slowly varying time functions are characterized by weight concentrated at low frequencies, while rapidly varying functions exhibit weight at higher frequencies [32]. The Fourier transform or series of a time function defines this *spectrum*.

Fig. 7 depicts the FFT values to conceive the results of the frequencies. Furthermore, the periodogram is exhibited, which computed from (10). The presented cases consist on the raw data and the differenced data as suggested from the previous results in order to obtain a stationary process.

From results of the application of the Power Spectrum Density (PSD) analysis in the differenced data, we observe a similar periodic component from the raw data. In order to prevent misleading results, we present the values of the periods from the FFT applied to differenced data where we visualize a significant peak in a low frequency value, close to 52 as shown in Table 4. Finally, the value belongs in the low frequency range, indicating that the time series expects to present a cycle regarding a long-term trend component. Therefore, the obtained value of the PSD demonstrates the dominant periodicity of the stationary process, and thus, the PED's seasonal component.

Spearman's Rank Correlation measures monotonicity of the relation between two variables, making it appropriate for capturing non-linear associations. The correlation assessment stated from Fig. 8 complements

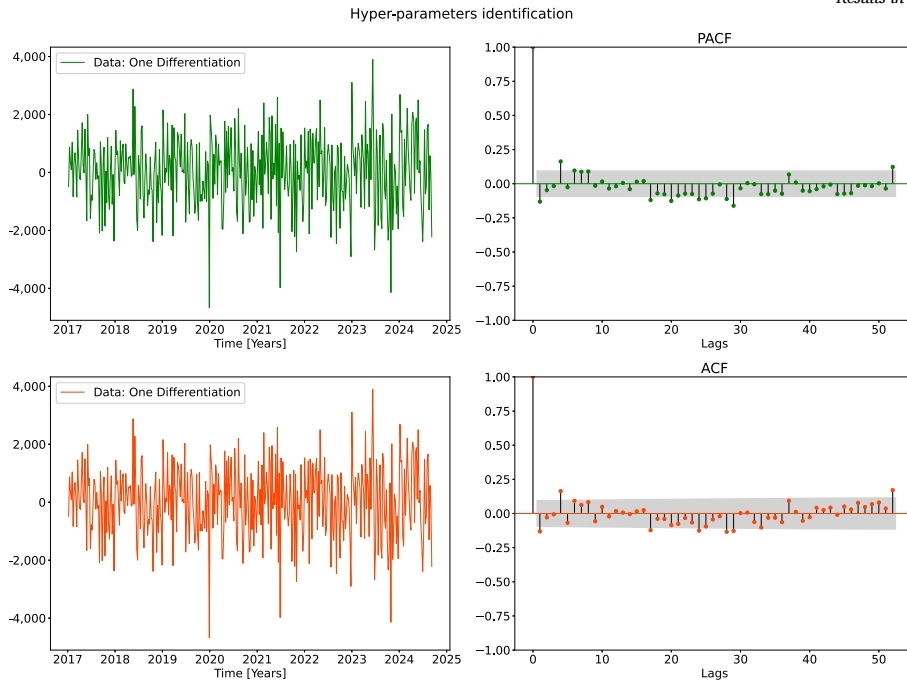


Fig. 6. Partial and Autocorrelation Function.

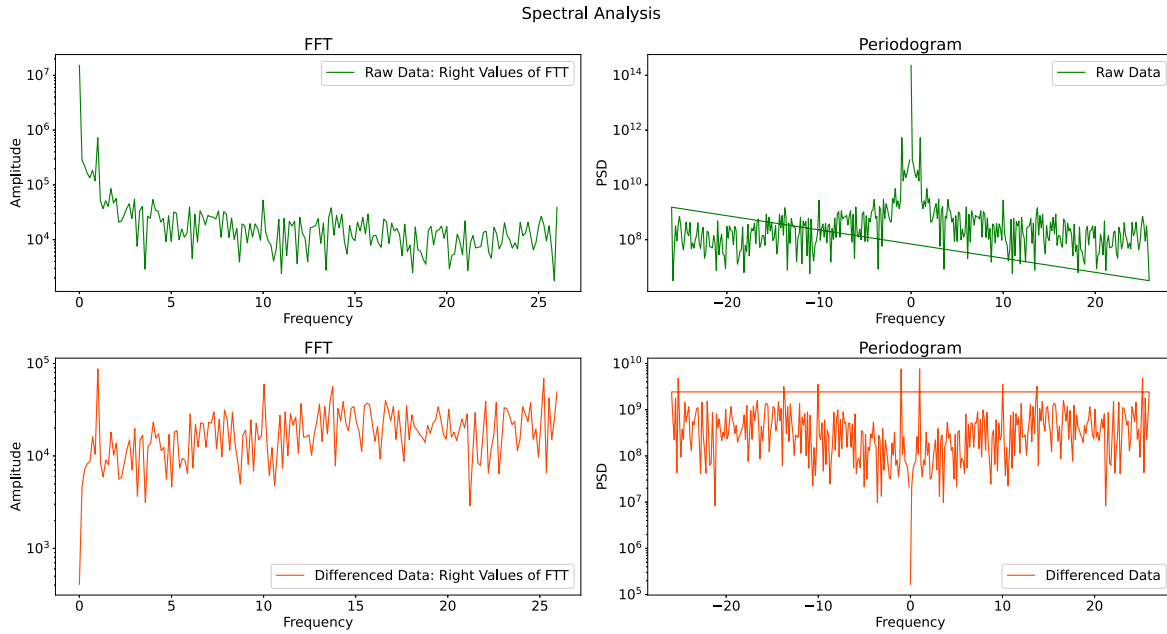


Fig. 7. Spectral Analysis on Raw and differenced data.

Table 4
Periods and frequencies with significant amplitude in the FFT plot.

Raw Data		Differenced Data	
Period	Frequency	Period	Frequency
52	0.019307	51.8571	0.0193
364	0.0027472	2.0625	0.4848
183	0.0054945	5.1857	0.1928

the time series decomposition by highlighting the relationships between temporal factors such year, week and PED.

Albeit the weak correlation between Week and Demand, it aligns with the presence of a seasonal component. Furthermore, to interact

with the seasonality, this paper introduces a normalized indicator of the week as an exogenous variable $\{X^e\}$, which represents the normalized cycle position index of each observation ranging from $(0, 1]$ and computed by $x_t^e = \frac{t}{52}$, $\forall t \in \{1, \dots, 52\}$.

Aiming to preserve parsimony and considering covariates that improve the model are not necessarily the most correlated ones, we include the normalized cycle position index as a linear factor with a resulting parameter of value equals to 0.3752.

The selection of the optimal SARIMA model was conducted through a structured and rigorous procedure, beginning with a preliminary inspection of the ACF and PACF to guide the initial choice of candidate orders. A systematic exploration was then carried out by incrementally adjusting the autoregressive parameter p , while holding other components

Table 5
Summary of estimated parameters of SARIMA model.

Parameter	Coefficient	Std. Error	Z-statistic	P > z	[0.025	0.975]
ar ₁	0.2468	0.151	1.638	0.101	-0.048	0.542
ma ₁	-0.5537	0.131	-4.218	0.000	-0.811	-0.296
AR ₅₂	-0.5213	0.051	-10.146	0.000	-0.622	-0.421
sigma ₂	1.765e+06	1.36e+05	12.976	0.000	1.5e+06	2.03e+06

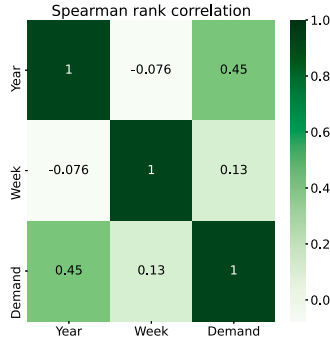


Fig. 8. Correlation between Input Variables.

fixed to maintain model parsimony. Competing models were evaluated using the AIC, with preference given to those exhibiting the lowest AIC values. Moreover, residual diagnostics, including spectral analysis via periodogram, were applied to ensure the adequacy of the residuals. The robustness of the final model was further supported by its superior forecasting accuracy on a hold-out validation set. Based on these combined criteria, the SARIMA(1,1,1) × (1,1,0)₅₂ specification was selected as the most appropriate structure for capturing the temporal dynamics of peak electricity demand in Mexico.

Table 5 summarizes the parameters of the SARIMA model this paper proposes. The parameters under consideration are the non-seasonal terms, autoregressive term ar₁ and the moving average term ma₁, both at lag 1, and the seasonal autoregressive term at lag AR₅₂.

Table 5 shows that the ar₁ coefficient has no reasonable impact on the model given its p-value equals 0.101, implying no statistical significance at 5% confidence intervals. The condition above is sustained by the confidence interval, which includes zero. Moreover, the dependence of past observations at lag 1 indicates a limited impact on the model's overall predictive capabilities due to the uncertainty of whether the effect would be positive or negative in the outcome. Nevertheless, when conducting a lag plot as shown in Fig. 9, temporal dependencies exist with a substantial correlation between the points and their lag at 1, contrary to lags 2 and 3, where the points tend to scatter. The association between each data point and its antecedent observation at lag 1 describes a plausible presence of autocorrelation. Consequently, given that the model minimizes the AIC values, we conduct the experiment by considering the coefficient in the modeling and evaluating possible future results.

On the other hand, the ma₁ coefficient highlights a statistical significance in the forecast due to its defined confidence intervals and p-value lower than 0.001. It reflects a strong dependency of past observations at lag 1, conversely from the ar₁ coefficient. The ma₁ parameter allows capturing the dataset's dynamics with an inverse relationship. Similarly, the AR₅₂ presents the condition the ma₁ coefficient has, with a p-value lower than 0.0001. The parameter captures the seasonal dependency at lag 52, as the spectral analysis supports it.

The obtained coefficients in the modeling have standard errors, which exhibit a significant degree of precision regarding forecasting. The Z-statistic shows the reliability mainly on the seasonal Autoregressive parameter coefficient, whose value is close to -10, indicating a statistical significance amongst the proposed parameters. Therefore, ma₁ and AR₅₂ have a discernible impact on the forecast outcome. Con-

Table 6
Residuals description from SARIMA model.

Parameter	Value
Count	400.000000
Mean	-4.196638
Std. Dev.	2401.921612
Min.	-16424.745159
25%	-937.714960
50%	40.474029
75%	774.372048
Max.	35961.992787

sequently, results from Table 5 describe the notable precision of the parameter estimates given the obtained standard errors and their confidence intervals. Moreover, the proposed parameters underscore the robustness of the proposed SARIMA model. Finally, the residual variance sigma₂, with a p-value lower than 0.001, further validates the model's robustness and ability to capture the inherent variability in the data. The residuals statistics from the fitted SARIMA model are shown in Table 6. From Table 6 and Fig. 10, we note that the residual mean is close to zero, meaning that it is considerably unbiased. Furthermore, we conducted the spectral analysis over the residual, as depicted in Fig. 11.

In the case of wide-sense stationary processes, the autocorrelation function $\hat{\rho}_X(h)$ effectively captures this average rate of change. A slowly varying random process retains self-correlation over longer time intervals, causing $\hat{\rho}_X(h)$ to decay gradually as the time lag h increases. Conversely, a rapidly fluctuating process loses self-correlation quickly, leading to a sharp decline in $\hat{\rho}_X(h)$ with increasing h [32].

Spectral analysis is a non-parametric procedure in which a finite set of observations is employed to estimate a function defined over the range $(0, \pi)$. Spectral analysis is at its most useful for series of the type with no obvious trend or seasonality, such as the White Noise process [31]. Therefore, we present an assessment based on spectral analysis to support the conducted statistical test on the times series, and to exploit the study in the frequency, the residuals from the fitted model are evaluated.

The properties in terms of autocovariance and autocorrelation function are displayed in order to highlight a specific characteristic of the White Noise. Let X_t be a sequence of uncorrelated random variables with zero mean and variance σ_X^2 , i.e., $X_t \sim WN(0, \sigma_X^2)$, then $\hat{\gamma}_X(0) = \sigma_X^2$ and $\hat{\gamma}_X(h) = 0$ for all $|h| > 0$. Furthermore, the autocorrelation $\hat{\rho}_X(0)$ is also σ_X^2 and its $\hat{\rho}_X(h) = 0$ for $h \neq 0$ [30,31]. Consequently:

$$S_X(\lambda) = \frac{\sigma_X^2}{2\pi}, \quad -\pi < \lambda < \pi \quad (11)$$

or similarly,

$$S_X(\lambda) = \sigma_X^2, \quad -\frac{1}{2} < \lambda < \frac{1}{2} \quad (12)$$

since $S_X(\lambda)$ is periodic in λ with period 1. Thus, the process has a flat PSD and contains all possible frequencies in equal measure. i.e., each frequency in the spectrum contributes equally to the variance of the process [30,32].

It is clear that the periodogram follows a relative flat spectrum indicating as close behavior of a White Noise process. Therefore, we demonstrate that the model has successfully pass the goodness-of-fit test, yielding to the validation of the model's parameterization its adequacy

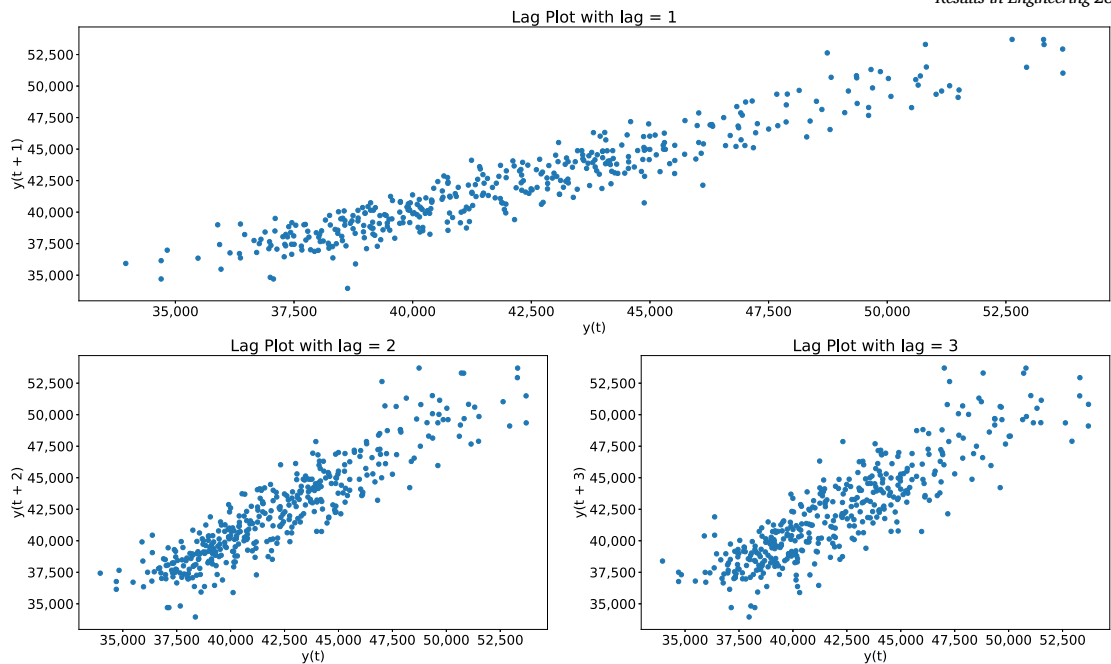


Fig. 9. Lag plots.

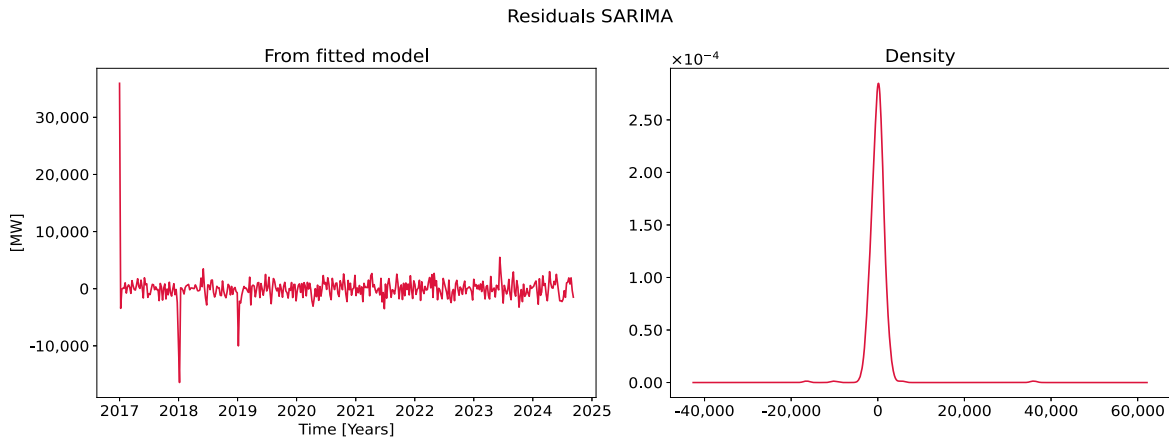


Fig. 10. Residuals obtained from the fitted SARIMA model.

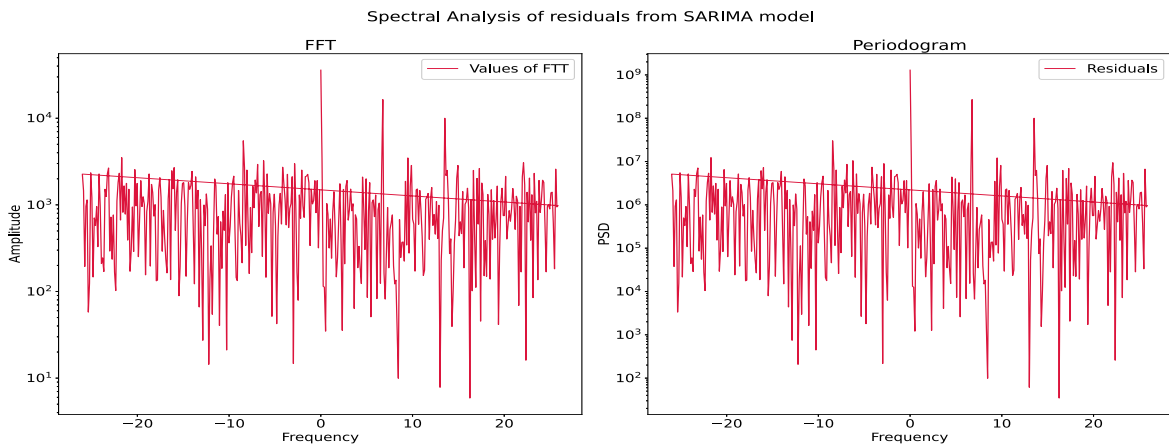


Fig. 11. Spectral Analysis on Residuals from fitted SARIMA model.

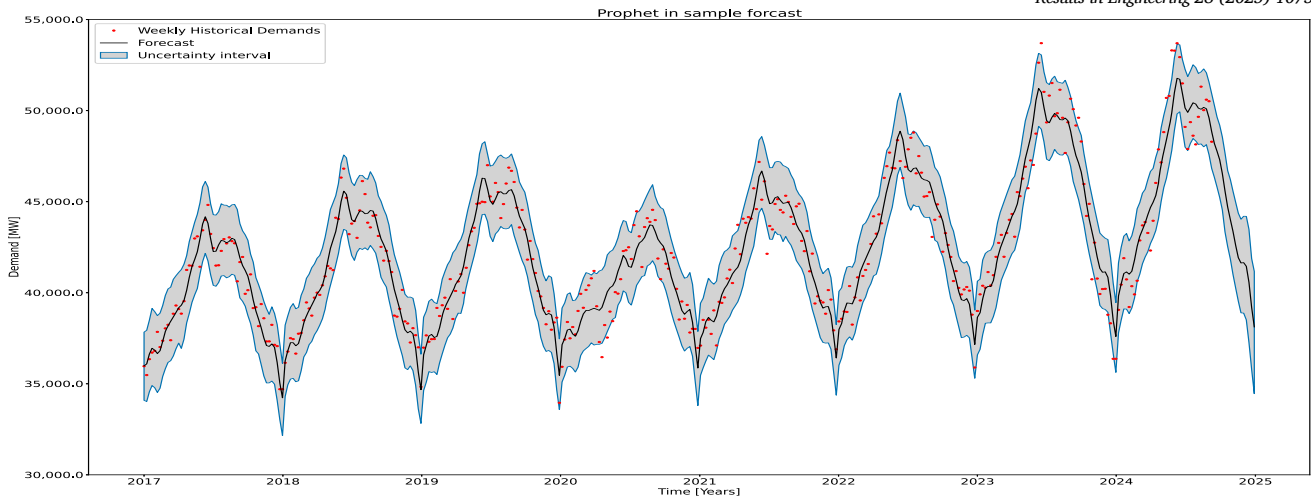


Fig. 12. Performance of Prophet for in sample forecasting.

to capture the information within M-SIN's PED. According to the flow process seen in Fig. 5, we shall stop and continue to forecast the values for unseen data.

4. Prophet model

The features to implement the Prophet model rely on a confidence interval of 95%, and due to non-linear seasonality, the model employs a yearly multiplicative component. Furthermore, using a regressor, we incorporate the cycle indicator feature in the model. The main hyper-parameter to define the flexibility and to capture shifts of the trend component in the model has a value of 0.6. It is worth mentioning that Prophet effectively models non-linear trends, seasonality, and holiday effects. Nevertheless, because the PED does not occur during holidays, the effect of seasonality is neglected. In Mexico, the maximum PED arises on either Wednesday or Thursday, and given that both days behave similarly, the model's performance might not improve if daily seasonality is considered.

4.1. Results

The results from the Prophet in sample forecast are shown in Fig. 12. The in sample predictions describe the suitability of Prophet to capture the multiplicative seasonality, tracing the seasonal fluctuations throughout the in sample period.

Results from Fig. 12 exhibit seasonal patterns under an annual periodicity. The observed behavior is consistent with the expected energy consumption regarding weather conditions. On the other hand, the uncertainty intervals incorporate the outline PED's variation, indicating a suitability to represent the model's uncertainty estimation. The heretofore capability of the model reflects a generalization of both seasonality and trend. Nevertheless, we further evaluate the over-fitting phenomenon over the model's performance concerning out of sample predictions.

Prophet is set to identify shifts in the trend component, Fig. 13, red dashed vertical lines. The changepoints from Fig. 13 are not arbitrarily placed; the model has adapted them to identify abrupt changes or reversals in demand fluctuations, reflecting alterations in the energy consumption patterns. The changepoints are uniformly placed in the first 80% of the historical time series data. When no changepoints are placed, we observe a substantial increment in the energy consumption. On the other hand, the red piece-wise line, representing the trend, tends to isolate structural shifts in long-term demand growth and provides insights for planning future operation and maintenance schedules.

The residuals statistics from the fitted Prophet model are shown in Table 7. Results shown in Table 7 conclude that the model is unbiased.

Table 7

Residuals description from Prophet model.

Parameter	Value
Count	400.000000
Mean	2.756056
Std. Dev.	1018.835596
Min.	-3251.454161
25%	-556.860835
50%	48.137124
75%	633.063813
Max.	3423.222910

Nevertheless, further analysis is carried out on the density plot shown in Fig. 14.

Fig. 14 depicts the residuals density plot with a skewed distribution. The aforementioned result could indicate that the model might not fit well, and further transformations on the data should be applied, such as a Box-Cox transformation.

Finally, to complete the analysis of the Prophet model, spectral analysis is conducted on the residuals as displayed in Fig. 15. Conversely, from SARIMA's residuals, the periodogram of Prophet's model is not entirely flat, suggesting that the model parameterization might need further improvement. Nonetheless, Prophet has a drawback in terms of parameterization, where there is limited control over some model aspects compared to traditional statistical models such as SARIMA [17]. Therefore, the parameter tuning is carried out as the models allow it.

5. XGBoost regressor model

The features implemented in the XGBoost regressor rely on the normalized cycle position index and the lags observed in the partial autocorrelation function. The former replaces the seasonal component from the seasonal decomposition of the original data; the residuals and trend are excluded in the modeling. On the other hand, from Fig. 6, we observe that the first four lags possible contain valuable information, since they are over the confidence interval. Therefore, four features for its respective shift are deployed in the training data.

Grid Search defines the hyper-parameter selection. Note that circumventing over-fitting could lead to under-fitting; consequently, we opt to tune only four parameters. The hyper-parameters to evaluate are the feature sub-sampling and the longest path from the root node to any leaf node in the decision tree, both to prevent over-fitting to specific features or noise in the training data. Furthermore, we evaluate the number of trees to be built in the ensemble to allow the model to generalize on unseen data and the learning rate to control the tree's contribution to the

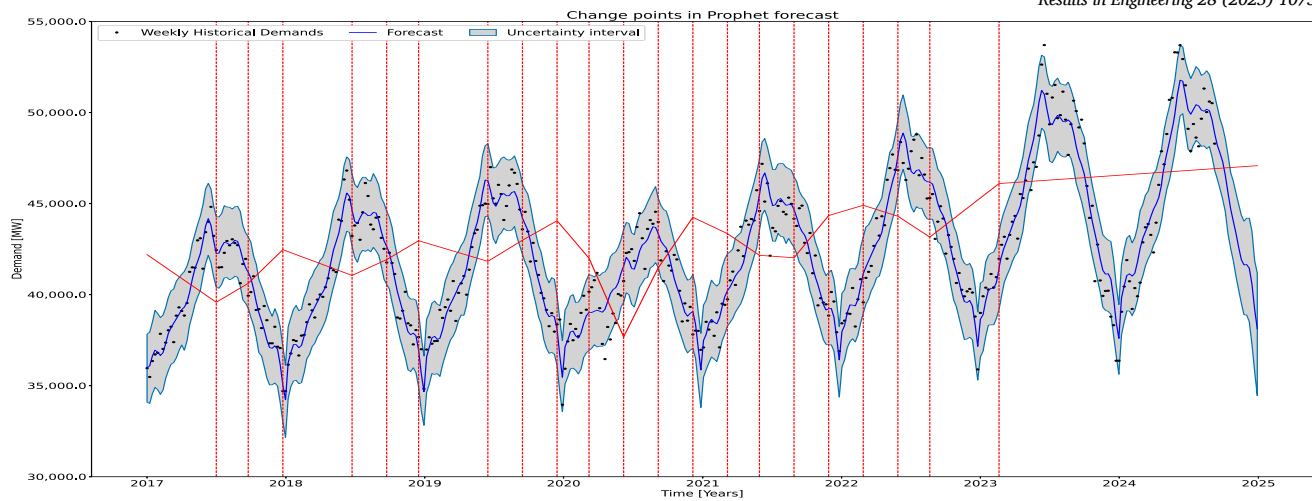


Fig. 13. Changepoints obtained from Prophet.

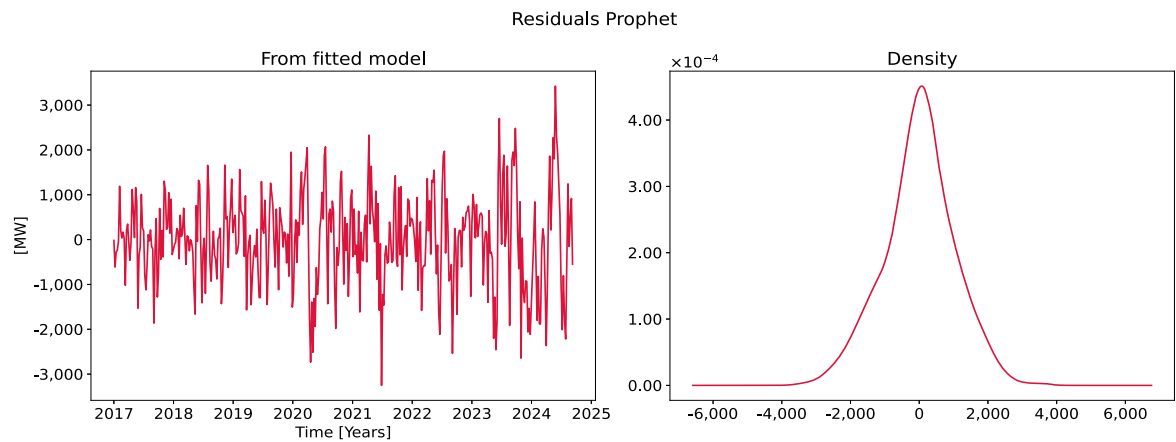


Fig. 14. Residuals obtained from the fitted Prophet model.

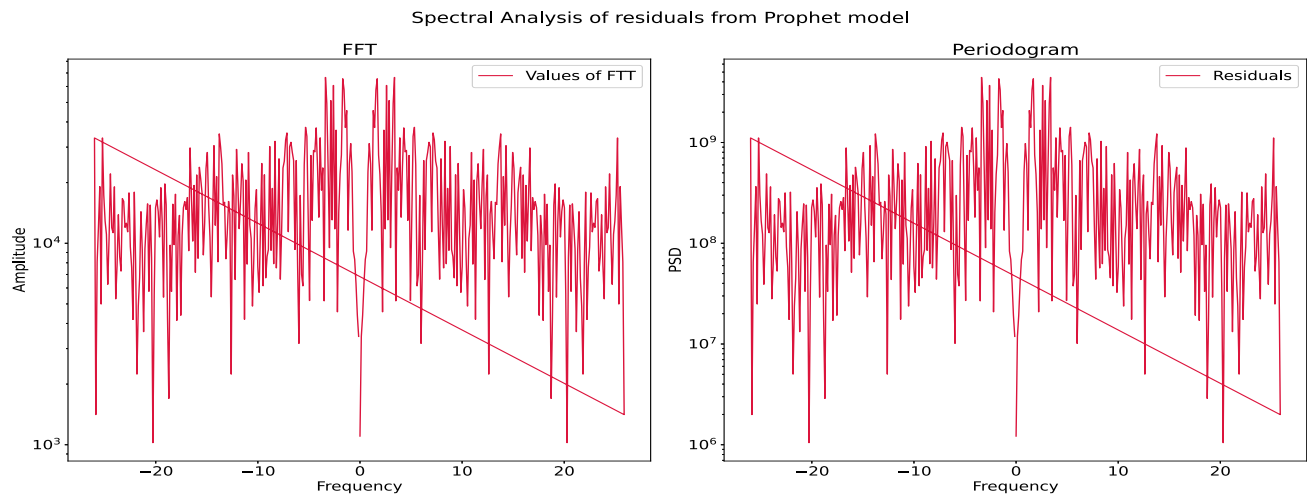


Fig. 15. Spectral Analysis on Residuals from fitted Prophet model.

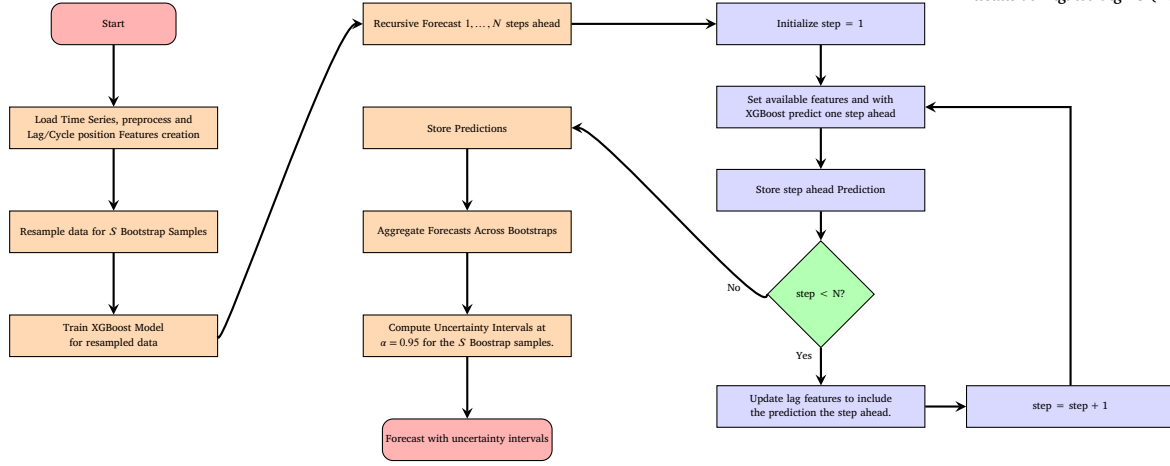


Fig. 16. Bootstrapped Recursive Forecasting.

Table 8
Residuals description from XGBoost model.

Parameter	Value
Count	400.000000
Mean	0.031246
Std. Dev.	825.271548
Min.	-2948.957031
25%	-498.146484
50%	23.917969
75%	575.067383
Max.	3423.222910

final prediction. The learning rate relates to the number of trees in the ensemble. The hyper-parameter tuning results in 0.7, 3, 500, and 0.01, respectively.

XGBoost does not provide uncertainty intervals directly, due to its deterministic nature, i.e., it is not a probabilistic machine learning model. Consequently, we opt to compute the uncertainty intervals based on the in sample forecast and the standard deviation from the residuals as presented below:

$$\begin{aligned} \text{Upper uncertainty interval} &= \hat{y}_t + 1.96\sigma \quad \forall t \in \{1, \dots, n\} \\ \text{Lower uncertainty interval} &= \hat{y}_t - 1.96\sigma \quad \forall t \in \{1, \dots, n\} \end{aligned} \quad (13)$$

where $\hat{y}_t$ is the one-step predictor of the actual value y_t for the n predicted values.

Moreover, XGBoost, a supervised machine learning model, requires features to predict future values, e.g., the lagged values. Similarly, the XGBoost does not provide the uncertainty interval for the forecast values. Therefore, the present paper introduces a bootstrapped recursive forecasting methodology, where predictions in S resampled data exist, at which each predicted value is an input for further predictions, similar to the workflow presented in [34]. The workflow for recursive forecasting is presented in Fig. 16.

5.1. Results

The results obtained from an in sample forecast using the XGBoost regressor are shown in Fig. 17. From predictions shown in Fig. 17, the model effectively learns and mimics the data variability, a feature that models such as exponential smoothing or ARIMA often underperform. The aforementioned suggests that XGBoost captures the seasonal structure of the data and the local dynamics, validating the lag and seasonal indicator. Nonetheless, unseen data analysis should be done to reject under or over-fitting. The residuals statistics from the fitted XGBoost model are shown in Table 8.

The residuals obtained from XGBoost regressor present the best statistics from the three models presented in the paper. The mean close to 0 supports that the forecast is unbiased. Results from Fig. 18 show the presence of skewness on the density plot, implying that the model requires further improvement in parameterization. Nonetheless, a forecasting method that does not satisfy these properties cannot necessarily be improved. Therefore, Grid Search performed on XGBoost regressor should vary more in parameters and consider other ones. Spectral analysis is conducted, and the results are shown in Fig. 19.

The periodogram obtained from XGBoost's residuals contradicts the observation made on the density plot. Conversely, the periodogram analysis validates the adequacy of the Grid Search parameters.

6. Forecast's performance

6.1. M-SIN dataset

To validate the predictions from the proposed model and with actual unseen data. First, the models are trained with values up to the 36th week of 2024, i.e., the first week of September. Second, the forecast is obtained towards the remaining weeks of 2024. Finally, the historical and forecast values disparity is shown in Fig. 20 for SARIMA, Prophet, and XGBoost models.

Prima facie from Fig. 20, the forecast from each model provides a valid prospect of the estimates. XGBoost validly estimates the probable future demands, albeit values close to week 52nd tend to diverge from SARIMA and Prophet models; further analysis on each forecast is presented. On the other hand, there is an expectation from data behavior that predictions tend to surpass 2023-PED's values due to a probable rise in temperature, industrial activity, and population. We visualize the aforementioned increase in the prediction from each model, indicating the adequacy to fit the time series trend for the Mexican PED.

The evaluation of the model's performance is conducted using all the data available, i.e., 400 recorded weeks, and then, the model generates future predictions of the remaining PEDs from 2024. Fig. 21, Fig. 22, and Fig. 23 depict both the recorded time series and the weekly PED values as time goes by toward the end of 2024 for SARIMA, Prophet, and XGBoost models, respectively.

From Fig. 21, one observes that the forecast captures the demand pattern. Furthermore, the uncertainty interval widens as the forecast extends further into the future, reflecting increased unreliability, typically immersed in time series forecasting due to the error's accumulation, indicating uncertainty in the forecast's precision to define future values. Regarding the forecast accuracy depending on the time horizon [35], the cumulative effect of prediction errors broadens as the time series expands for the near term of 2024. The model captures the general trend

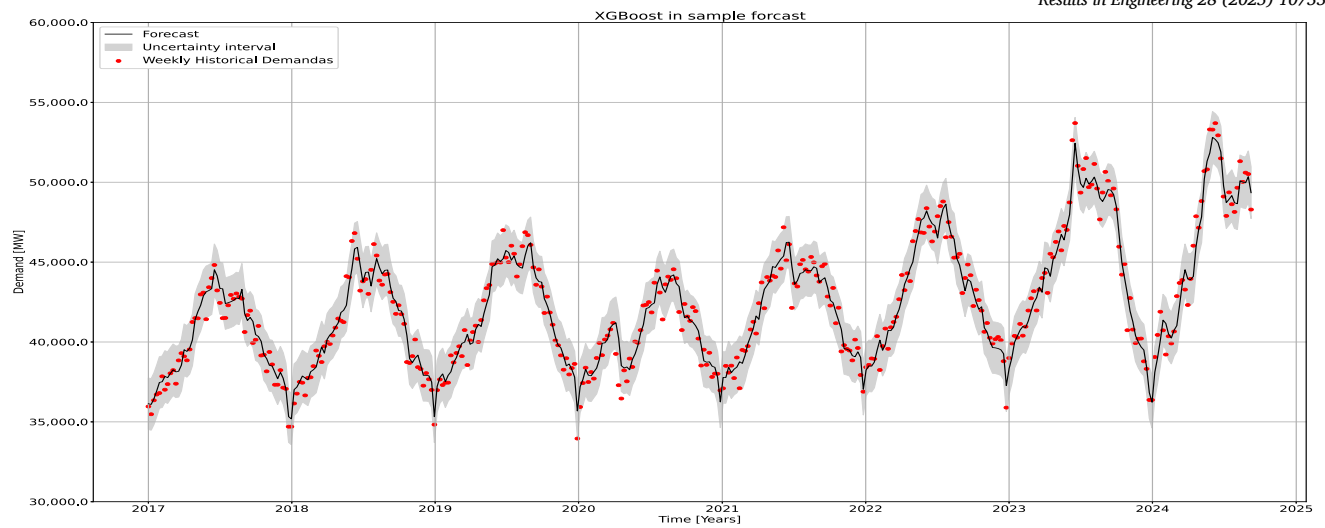


Fig. 17. Performance of XGBoost for in sample forecasting.

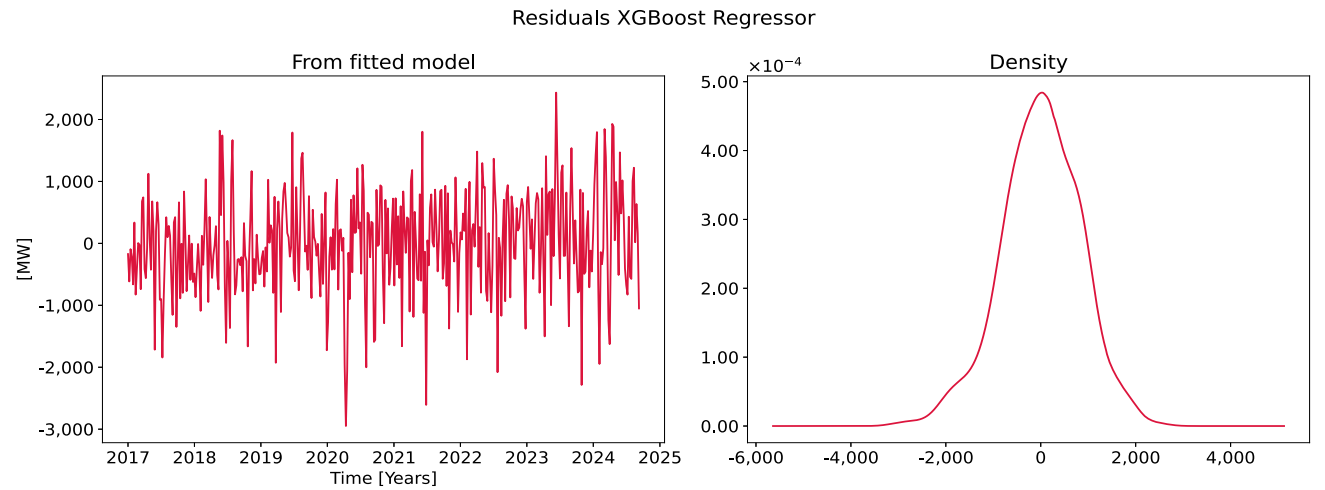


Fig. 18. Residuals obtained from the fitted XGBoost model.

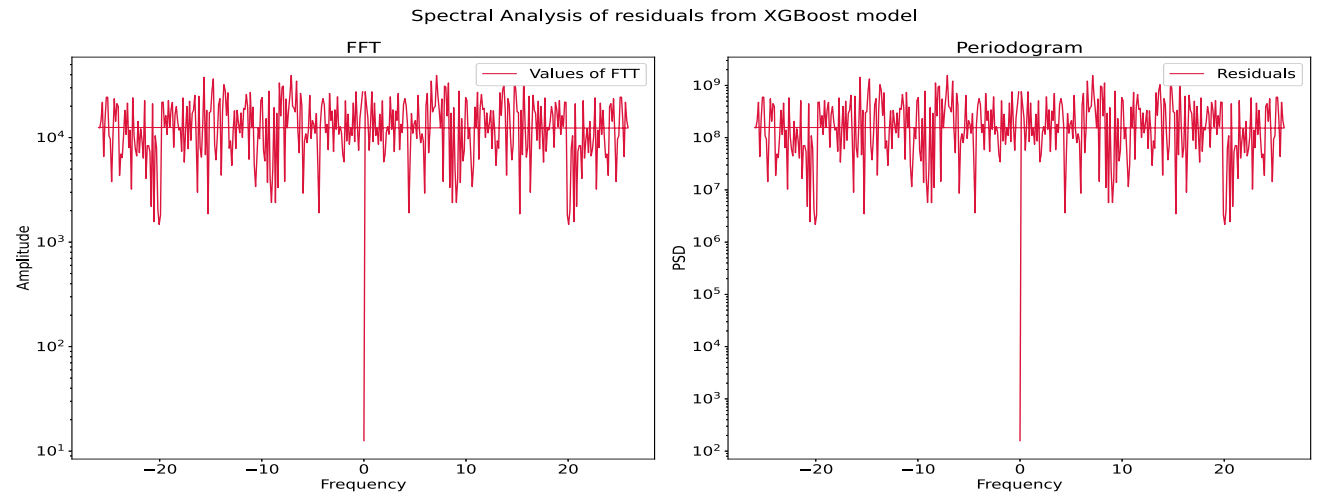


Fig. 19. Spectral Analysis on Residuals from fitted XGBoost model.

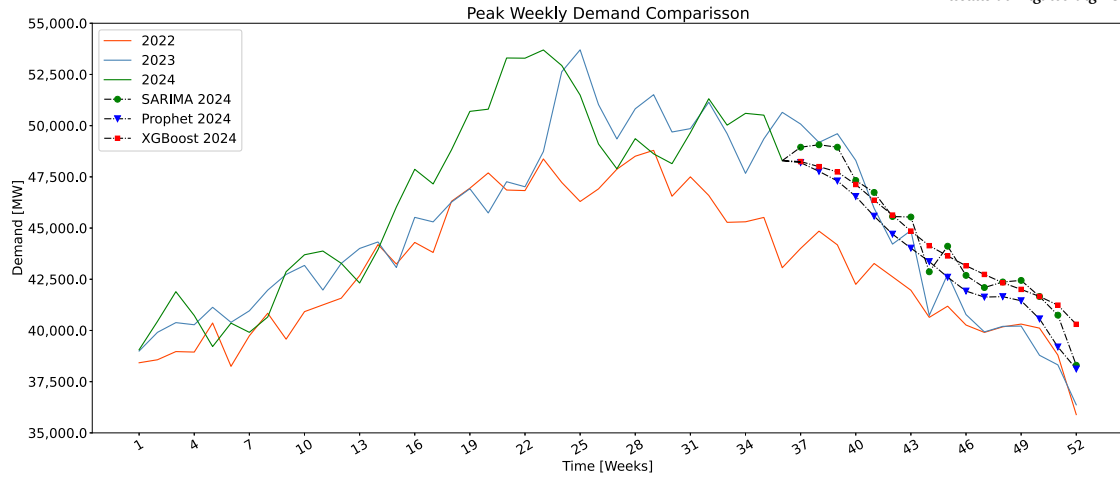


Fig. 20. Comparative Forecast with historical data.

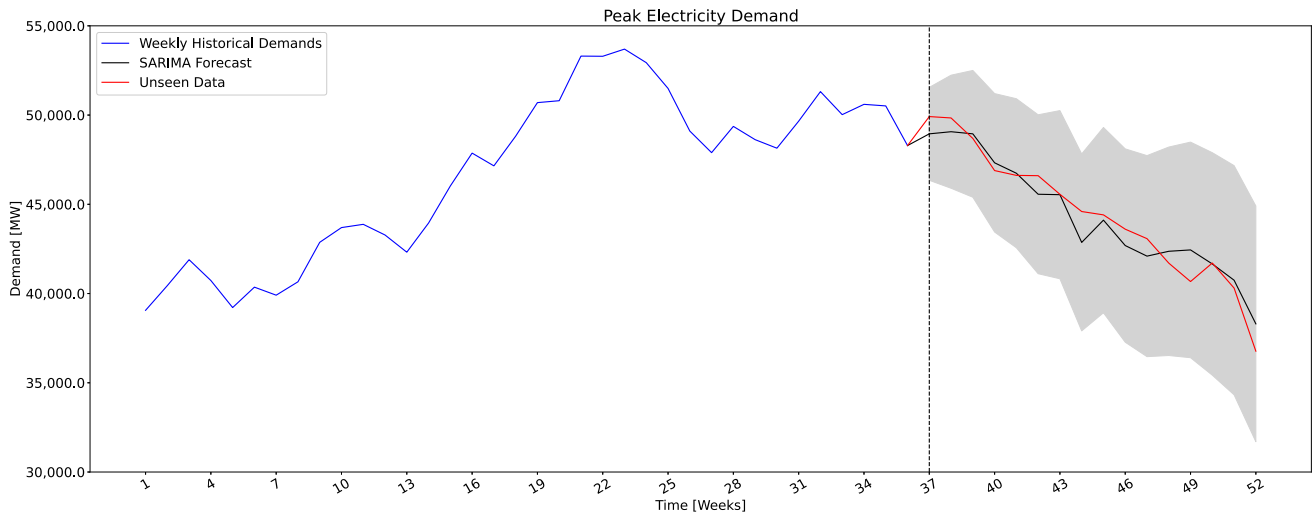


Fig. 21. Future PED Trajectory from SARIMA.

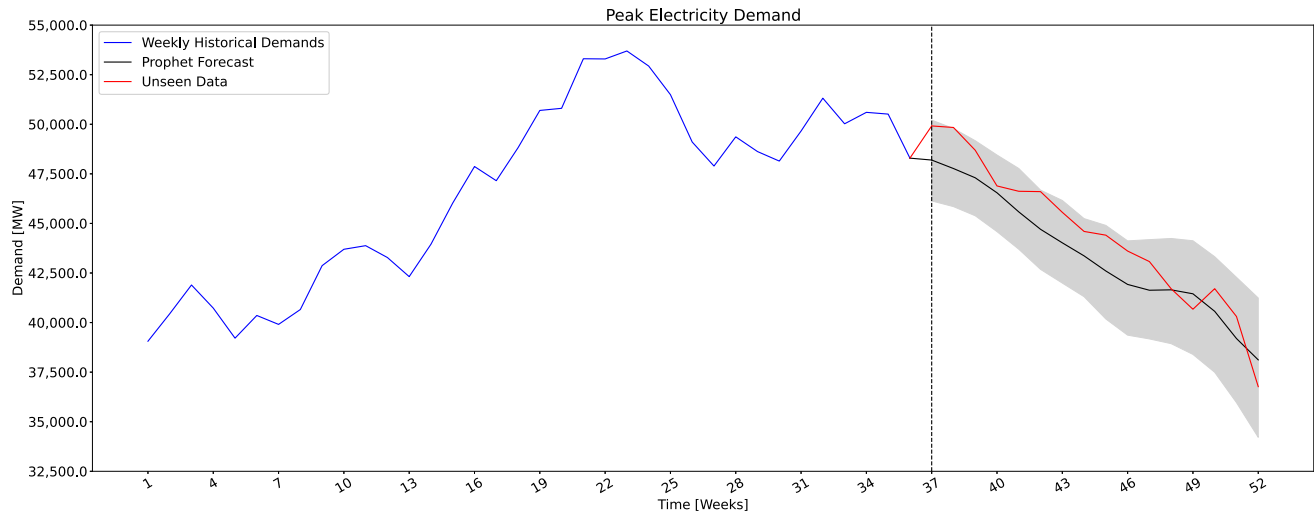


Fig. 22. Future PED Trajectory from Prophet.

and seasonal patterns. However, in certain instances, the value is underfitted. The implication above is due to the influence of the atypical values presented in 2020-2021. Therefore, the results influence performance due to its dependency on outliers and deviations.

We observe from Fig. 22 that Prophet out-of-sample prediction cannot mimic the variability of the demand over unseen data. As reported in [17], Prophet may not capture short-term patterns or sudden changes as effectively as other models, i.e., Prophet may lead to underfitting in

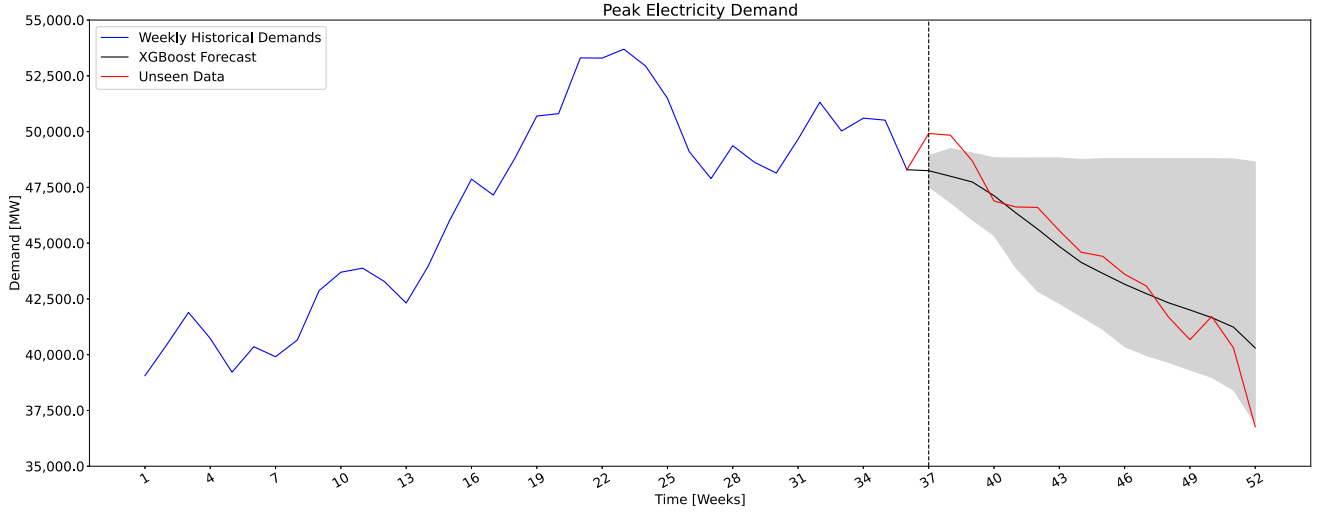


Fig. 23. Future PED Trajectory from XGBoost.

complex or high-stakes scenarios. The forecasting behavior could indicate that further analysis of Fourier orders on the seasonality parameter will be carried out. Nevertheless, it is worth mentioning that indicating high values of the Fourier order could compromise, at some point, the seasonalities. Furthermore, a negative bias exists in the long-term forecasts, leading to an underestimation of future values, which may be present even after parameter tuning due to Prophet characteristics.

On the other hand, Prophet excels over SARIMA and XGBoost models by providing narrow certainty ranges in future instances. We observe that the cumulative error does not broaden the uncertainty interval as much as SARIMA model. Therefore, the model's uncertainty on seasonality adapts correctly to unseen data, compromising the model's analysis of the Fourier orders. The aforementioned contradiction raises the question of the extent to which a beneficial tune Fourier-orders parameter? Is it worth risking certainty over adaptation? The answer to those questions could be based on the decision-making process that M-SIN's operators use. Deliberate decisions imply providing security conditions and thoughtful alternatives to satisfy the system requirements. Consequently, we opt to define a model that implies certainty over a model that improves its adaptation.

XGBoost predictions shown in Fig. 23 describe a similar behavior as Prophet. Nevertheless, the uncertainty intervals are compromised. The upper interval obtained from the Bootstrap algorithm tends to broaden significantly as we increase the number of steps ahead to consider. We observe that the lower one captures correctly the trend, as it tends to fall below the unseen data. It is worth noting that, although XGBoost has been the model that fits the data best, there is a drawback to the certainty of the obtained predictions. The results in Fig. 23 from the XGBoost model align with what was previously mentioned in Fig. 20 at the end of the planning span.

The measurement of error metrics completes the overall analysis of the forecast performance. We employ the Mean Absolute Error (MAE) to assess the deviation of the predictions from the actual values on average, the Root Mean Squared Error (RMSE) to evaluate results while penalizing large errors, the Mean Absolute Percentage Error (MAPE) to examine the ratio of the average errors between the actual and predicted values. Moreover, we apply the Mean Absolute Scaled Error (MASE) to interpret the forecast accuracy, considering the absolute errors concerning the in-sample mean absolute error from a naive forecast method. Finally, the R^2 is employed to provide information about the goodness of fit for each model. These metric errors provide us with actionable insights into the accuracy of our forecasts and are defined as follows:

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad (14)$$

Table 9

Error metrics description.

Metric	SARIMA	Prophet	XGBoost
MAE	749.6953[MW]	1287.9027[MW]	944.5286[MW]
RMSE	931.82[MW]	1392.56[MW]	1256.40[MW]
MAPE	1.7333[%]	2.8832[%]	2.1910[%]
MASE	0.7387[%]	1.2691[%]	0.9307[%]
R^2	0.9311	0.8460	0.8747

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (15)$$

$$MAPE = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right| \quad (16)$$

$$MASE = \frac{100}{n} \sum_{t=1}^n \frac{|y_t - \hat{y}_t|}{\frac{1}{n-1} \sum_{t=2}^n |y_t - y_{t-1}|} \quad (17)$$

$$R^2 = 1 - \frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{\sum_{t=1}^n (y_t - \bar{y})^2} \quad (18)$$

where $\hat{y}_t$ is the one-step predictor of the actual value y_t for the n predicted values, and $\bar{y}$ is the mean of the observed data.

Table 9 shows a comparative evaluation of the performance of SARIMA, Prophet, and XGBoost predictions.

To contextualize the performance of the forecasting models, we implement a seasonal naïve baseline, where the prediction for each week is set equal to the observed value from the same week in the previous year. It is worth mentioning that MASE compares the proposed forecast against an in sample naïve forecast. Essentially, having a MASE of 2.1159[%] implies that the seasonal naïve forecast is, approximately on average, 111.59[%] worse at predicting the PED than simply using the previous value as the prediction of the current demand. Furthermore, as suggested by [36], the MAPE of 4.9218[%] of the seasonal naïve forecast satisfies the condition of being less than 10[%] and thus, "highly accurate", which is debatable due to MASE error. Finally, the seasonal naïve baseline yielded an MAE of 2147.2667[MW] and an RMSE of 2560.74[MW], significant values regarding maximum demand. The predictions under analysis correspond to demands that tend to reduce as time progresses towards the end of the year. Thus, obtaining values above 2[GW] implies substantial deviation from the real data. These results become a point of departure to confirm that each model sig-

nificantly outperforms basic benchmark approaches and validates its suitability for operational deployment in PED forecasting for the M-SIN.

From Table 9, we obtain an average deviation of the SARIMA's predictions by approximately 750 [MW] from the actual values. The RMSE aligns with the MAE value, with an average 932 [MW] deviation, by placing greater weight on large errors. The particularity of small MAPE reinforces its robust predictive capability. Furthermore, the MASE being lower than 1 reaffirms the model's superiority over naive benchmarks and its reliability for time series forecasting.

MAE, RMSE, and MAPE errors from XGBoost outperform those obtained from Prophet's model, and fall behind those from SARIMA's model. XGBoost's performance excels in its utility when there is sufficient data and lagged features, maintaining reasonable generalization. Its MASE and RMSE indicate its adequacy for operational deployment, mainly on scenarios affected by nonlinear factors. It is worth mentioning that the MASE error falls below the threshold of 1, confirming the model's effectiveness compared to simplistic models.

Prophet's predictions yield the highest MAE and RMSE amongst the three proposed models. The aforementioned errors describe the model's difficulty in capturing the data's variability, despite its ability to detect seasonality automatically and decompose the trend. Furthermore, Prophet predictions obtain an MASE bigger than 1, suggesting that the model is not capturing the underlying patterns in the data. Prophet is an advanced time series forecasting procedure. Nevertheless, its performance might be compromised if data presents either weak seasonality or substantial changes that the model has not yet foreseen. Therefore, Prophet's model may not be the best choice for the time series employed in this paper.

Finally, the coefficient of determination R^2 describes that the SARIMA model captures 93.11% of the time series variance. On the other hand, Prophet underperforms concerning SARIMA and XGBoost, which may result from the simplifications considered on the seasonality decomposition. According to the aforementioned errors, Prophet has been unable to reproduce short-term dependencies. Moreover, XGBoost implies an ability to fit non-linearities even in the presence of complex seasonalities. Regarding metrics deployed in Table 9, one can observe that the proposed SARIMA model demonstrates its capability to predict accurately.

The computer used to run the simulations regarding the training times is a 64-bit Intel(R) Core(TM) i7 - 1165G7 CPU 2.80 [GHz] x4 cores with Windows 10 and 12 GB of RAM at 3200 [MHz]. To obtain the statistics for each model, the training stage is executed 100 times to foresee how the execution time might vary across different runs due to background processes and system load. Fig. 24 depicts the distribution and statistics of the elapsed time for each model, where the right vertical axis corresponds to SARIMA, and the left corresponds to XGBoost and Prophet.

The statistics in Fig. 24 reflect the inherent computational characteristics of each model. SARIMA requires iterative maximum likelihood estimation over multiple seasonal and non-seasonal parameters, resulting in longer training times with a mean of 20.97 [seconds]. In contrast, Prophet detaches with a mean of 0.30 [seconds], and XGBoost, with a mean of approximately 0.47 [seconds]. XGBoost and Prophet converge almost instantly in terms of their algorithmic efficiency. These results indicate that XGBoost and Prophet provide highly stable and predictable training performance, while SARIMA describes greater variability and longer computation times. Therefore, the observed differences in training times directly highlight the superior efficiency of the proposed approaches.

6.2. M-BCS dataset

The following evaluation is based on the time series of the PED in the M-BCS system [28], where the data recorded comprehends from January, 2017 to the first week of September, 2024, as depicted in Fig. 25.

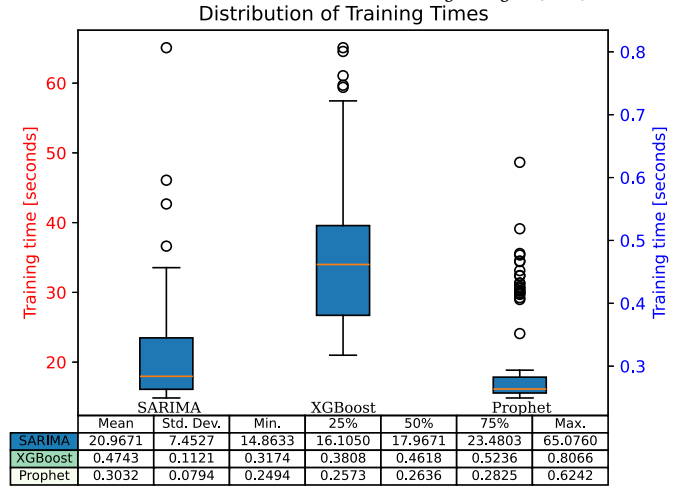


Fig. 24. Statistics and distribution of training times for each model.

The SARIMA model specified for the M-BCS dataset adopts a structure analogous to that used for the M-SIN dataset: $SARIMA(2,0,1) \times (1,1,0)_{52}$. The non-seasonal differencing order was set to $d = 0$ based on statistical tests confirming stationarity in the mean, while seasonal differencing was set to $D = 1$ to stabilize the annual seasonal component. This choice is justified by the observed yearly repetition in demand patterns and the relatively stable demand level, which aligns with the region's low population growth. The autoregressive order $p = 2$ was selected after evaluating model errors and minimizing the AIC. Furthermore, given the geographical proximity of the M-BCS and M-SIN regions, their climatic conditions are comparable, resulting in similar seasonal behaviors.

For the Prophet model, the flexibility parameter was set to 0.5 to balance under and overfitting, and yearly seasonality was enabled to capture annual periodic effects. In the case of the XGBoost model, five lagged features were included based on ACF and PACF analyses. Hyperparameter tuning yielded an optimal configuration of 1,000 estimators, a learning rate of 0.001, a subsample ratio of 0.7, and a maximum tree depth of 3. These models were applied to the M-BCS dataset using the same methodological framework and selection criteria as described previously. The resulting performance supports the consistency, robustness, and generalization capability of the proposed approach across varying regional contexts.

The comparative visualization of 2024-PED, alongside historical data from 2017 and 2024, shown in Fig. 26, reveals distinct behavioral characteristics across the three forecasting models. SARIMA demonstrates a strong balance between performance and interpretability, highlighting its ability to mimic seasonal patterns with minimal deviation as seen in the 2024 projections.

The SARIMA model exhibits a smooth and consistent forecast trajectory that closely aligns with the seasonal patterns from the M-BCS data as depicted in Fig. 27.

As illustrated in Fig. 27, the SARIMA model effectively replicates the cyclical behavior of weekly PED, capturing both the amplitude and phase of recurring seasonal patterns with notable precision. The model demonstrates strong alignment with historical data, particularly in terms of long-term seasonality and autocorrelation structure, thereby reinforcing its robustness and reliability for forecasting applications. This characteristic is especially beneficial in operational settings where accurate demand planning requires stable, interpretable, and seasonally consistent outputs.

In contrast, the Prophet model, shown in Fig. 28, exhibits reduced coherence with the timing and magnitude of historical seasonal peaks. Although the model successfully captures the overall trend, its forecast intervals fail to encompass the observed values during certain periods, particularly around weeks 43 and 44, highlighting potential limitations

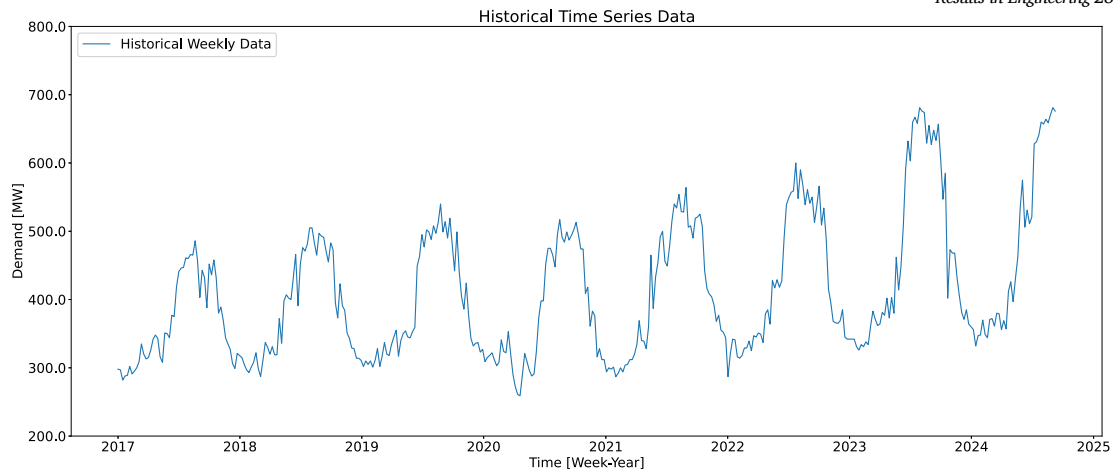


Fig. 25. Second Raw Dataset.

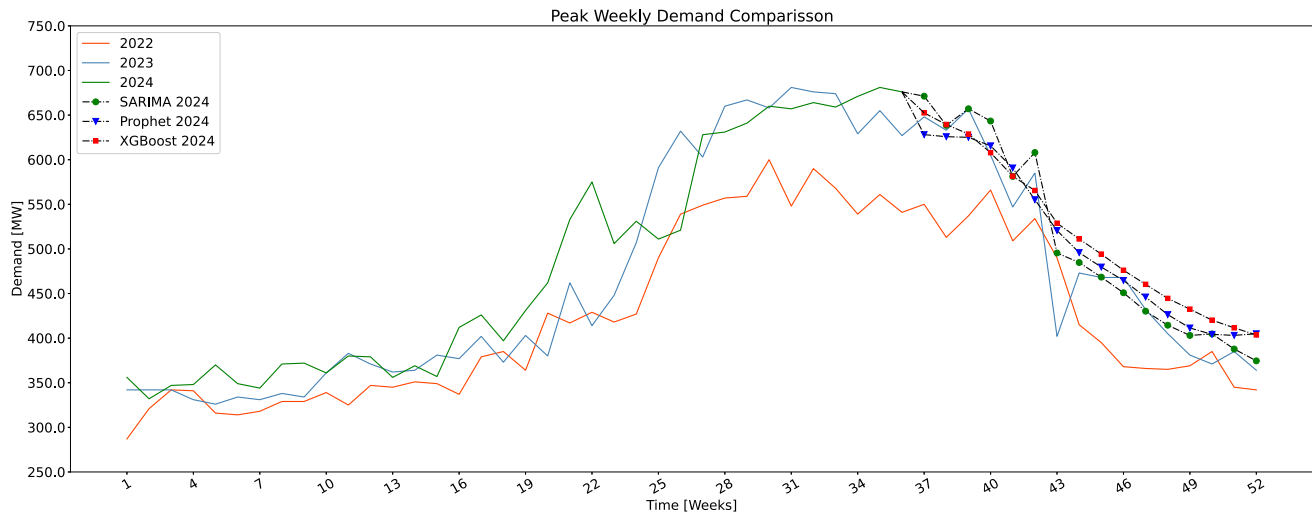


Fig. 26. Comparative Forecast with historical data from validation set.

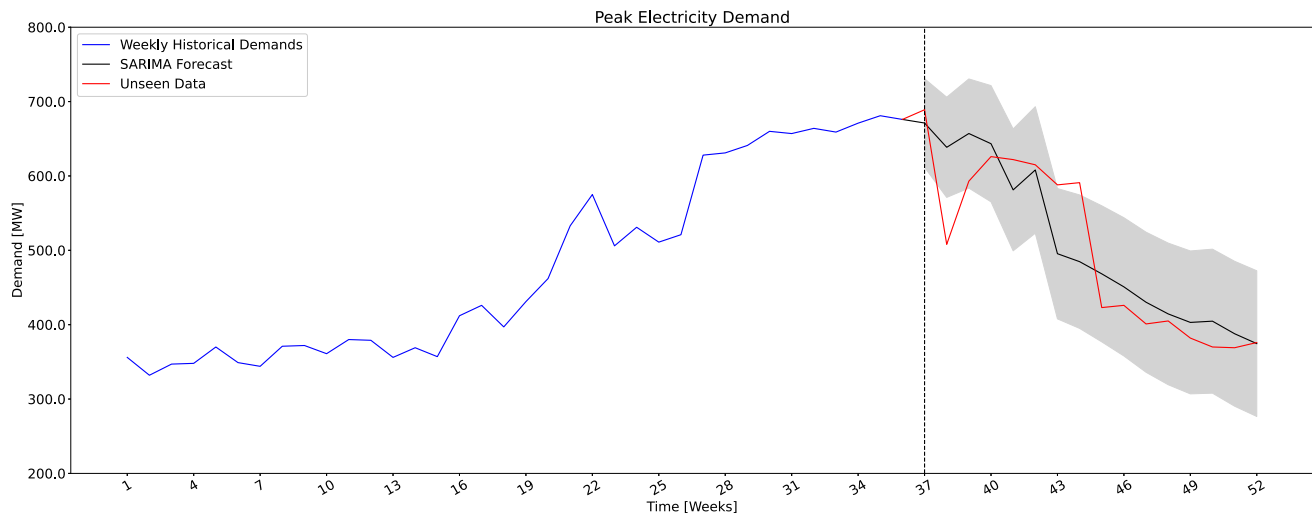


Fig. 27. Future PED Trajectory from SARIMA on second dataset.

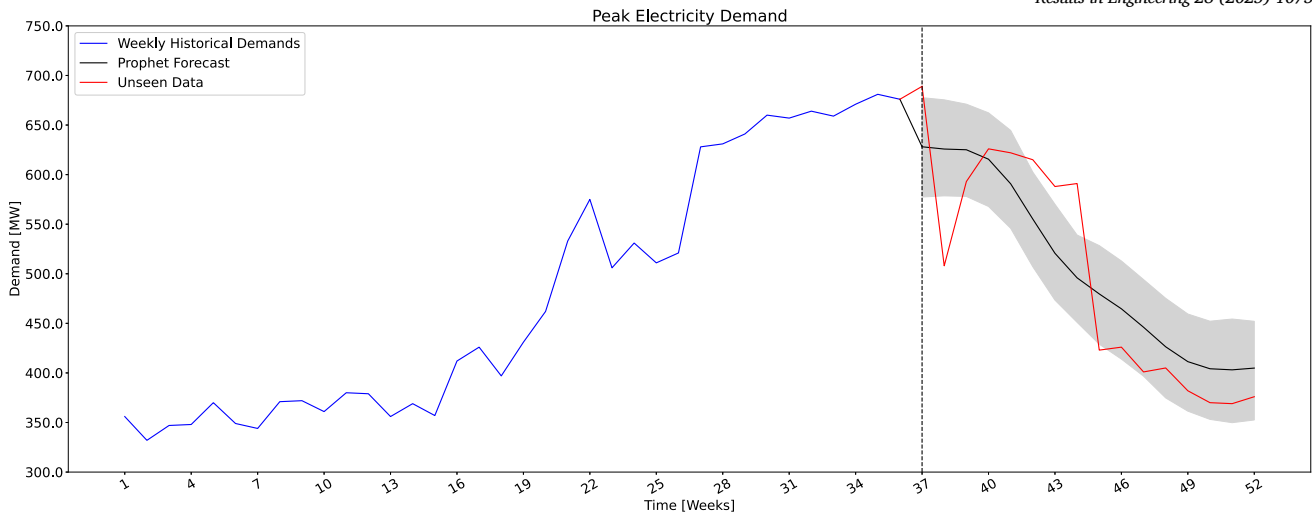


Fig. 28. Future PED Trajectory from Prophet on second dataset.

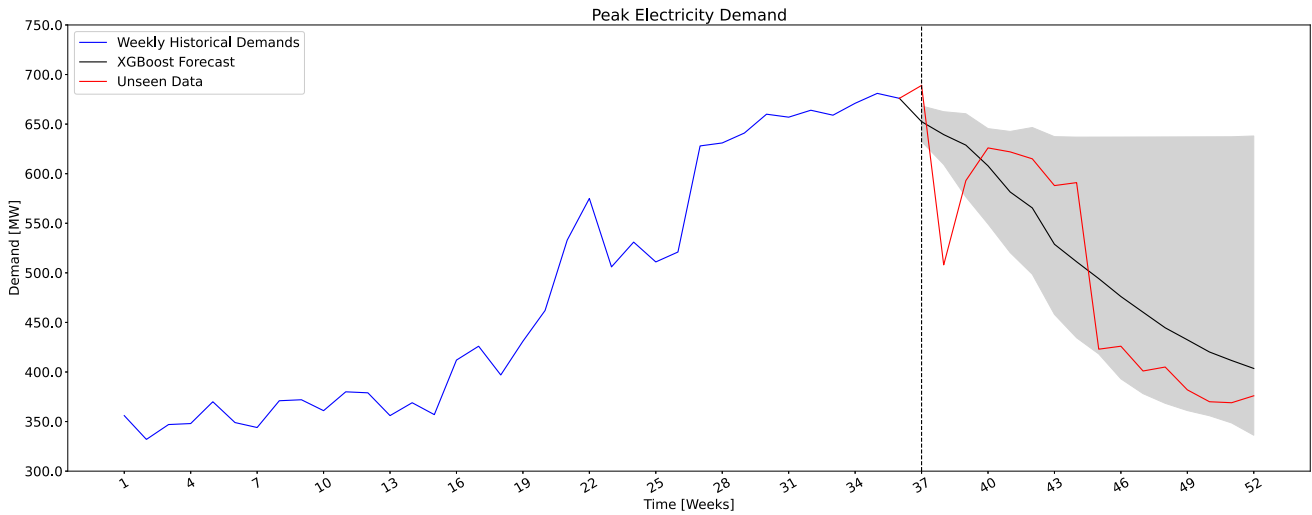


Fig. 29. Future PED Trajectory from XGBoost on second dataset.

in the changepoint detection mechanism or the seasonal component estimation when applied to PED time series.

Meanwhile, the XGBoost model Fig. 29 produces smoother forecast trajectories with relatively low short-term variability. While this approach may suppress some of the finer seasonal fluctuations, its stable forecasting behavior suggests suitability for applications prioritizing conservative or smoothed demand estimations.

The XGBoost's behavior depicted in Fig. 29, reflects higher uncertainty when performing a forecast. This highlights the model's sensitivity to short-term variations and noise in the training data. Consequently, this characteristic may compromise the model's generalization ability across seasonal cycles.

The forecasting error metrics obtained for the M-BCS dataset are generally consistent with those reported for the M-SIN region, underscoring the generalizability of the proposed modeling framework across geographically and operationally similar systems. This consistency suggests that the methodology does not overfit to a particular instance, and can be reliably extended to regions with comparable demand characteristics. A comparative summary of model performance across SARIMA, Prophet, and XGBoost is provided in Table 10.

The seasonal naïve benchmark yields a MASE of 1.4569, indicating that its predictions are, on average, approximately 45.69[%] less accurate than a naïve forecast using the most recent observation. Additionally, the seasonal naïve model records a MAPE of 10.2191[%], an

Table 10

Error metrics description for the M-BCS dataset.

Metric	SARIMA	Prophet	XGBoost
MAE	41.3283 [MW]	47.6955 [MW]	52.6102 [MW]
RMSE	55.31 [MW]	54.82 [MW]	58.35 [MW]
MAPE	8.1143 [%]	9.6411 [%]	11.0636 [%]
MASE	1.0633 [%]	1.2272 [%]	1.3536 [%]
R^2	0.7528	0.7572	0.7249

MAE of 56.6250 [MW], and an RMSE of 82.91 MW[MW]. These metrics serve as a baseline, against which all three advanced models demonstrate improved performance, with the exception of XGBoost under the MAPE criterion.

Among the candidate models, Prophet achieves the lowest RMSE and the highest $R^2 = 0.7572$, reflecting its strong fit to the validation data. However, the SARIMA model remains a highly competitive and arguably more practical alternative, delivering robust MAE, MAPE, and MASE values alongside a strong R^2 of 0.7528 and a markedly lower RMSE of 55.31 [MW] compared to the baseline. Although XGBoost exhibits proficiency in terms of some error metrics, likely due to its ensemble learning structure, its performance is less stable in this context and may require further tuning for optimal deployment.

From a methodological perspective, SARIMA provides significant advantages in terms of statistical transparency, parameter interpretability,

Table 11
Comparative Summary of Forecasting Models.

Aspect	SARIMA	Prophet	XGBoost
Model Type	Statistical	Additive time series decomposition	Machine learning (ensemble trees)
Input Features	Historical PED + cycle index	Historical PED + cycle index	Lags 4–5 + cycle index
Seasonality Handling	AR ₅₂ term	Automatic detection (Fourier series)	Lagged features + cycle index
Cycle Representation	Normalized index (exogenous)	Regressor (continuous)	Regressor (continuous)
Nonlinearity Modeling	Limited (linear)	Piecewise trend flexibility	Fully nonlinear with trees
Hyperparameter Tuning	AIC + spectral analysis	Changepoints + seasonality prior + spectral analysis	Grid search + spectral analysis
Interpretability	High (coefficients available)	Moderate (trend, changepoints)	Low (requires feature importance analysis)
Uncertainty Quantification	Parametric confidence intervals	Built-in Bayesian intervals	Bootstrapped intervals (empirical)
Computational Complexity	High	Moderate	Moderate
Strengths	Interpretable, effective for seasonality	Captures structural changes	Handles nonlinearity and variance
Limitations	Assumes stationarity, sensitive to outliers	May miss high-frequency variation	Needs tuning, lacks native uncertainty quantification

and explicit seasonality modeling. These features are especially relevant in operational and regulatory environments where model explainability and traceability are essential. Unlike black-box machine learning models, SARIMA allows for structured diagnostic analysis, facilitates uncertainty quantification, and aligns naturally with the recurrent patterns typical of Mexico's electricity demand profiles.

Overall, while Prophet slightly outperforms SARIMA in certain statistical measures, SARIMA offers a superior balance between forecasting accuracy, structural consistency, and interpretability. Its transparent formulation and strong alignment with historical seasonal behaviors position it as a technically sound and operationally reliable choice for peak electricity demand forecasting in the Mexican context.

Finally, according to the obtained results, we deploy a comparative summary of the implemented forecasting models in Table 11.

7. Conclusions

This study underscores the critical role of electricity demand forecasting in the long-term planning of the Mexican power system. Accurate forecasts are indispensable for mitigating operational delays and improving system reliability. They are essential for ensuring coordinated development between generation expansion and transmission infrastructure. The Mexican Independent System Operator already integrates peak demand forecasts into operational planning. This occurs notably through optimizing maintenance schedules based on deterministic load profiles. These schedules directly influence system security, reliability, and reserve margin calculations. Consequently, the predictive accuracy of PED becomes a foundational element upon which broader operational decisions are based.

In contrast to existing studies that often rely solely on either statistical or machine learning techniques, this work presents a comparative framework. We integrate classical time series models, such as SARIMA, with modern approaches such as Prophet and XGBoost. By evaluating model performance across various forecast horizons, error structures, and seasonal patterns, we provide a comprehensive understanding of each model's capabilities and limitations. Our findings are further reinforced by applying the models to the M-SIN dataset and the M-BCS

region. The latter is a geographically distinct and underexplored area with unique demand characteristics. This transferability demonstrates the generalizability of our framework. Moreover, it enhances practical relevance for energy systems with similar data constraints.

The integration of a normalized index, alongside spectral analysis, strengthens the methodological foundation of our approach. The models collectively provide a reliable characterization of the expected weekly PED profile for 2024, offering actionable insights for generation planning and maintenance scheduling. While the spectral analysis does not directly influence forecast accuracy, it validates the model's temporal structure and supports the selection of hyperparameters in a statistically coherent manner.

Future research should explore the integration of exogenous variables, such as precipitation, humidity, wind speed, cloud cover, and reserve requirements, to capture potential nonlinearities and improve forecast fidelity. In the Mexican context, including operative requirement ratios may prove valuable for refining operational forecasting. Additionally, enhancements to spectral diagnostics, such as the Welch method, could improve frequency resolution and reduce variance, offering more profound insights into seasonal behavior.

This work establishes a replicable and robust forecasting methodology with significant and immediate implications for Mexico's electricity system operation and planning. The results demonstrate the superiority of hybrid model assessment frameworks in policy-relevant energy modeling. This research shows that prioritizing interpretability, accuracy, and generalizability in demand forecasting is essential for advancing effective, data-driven planning and decision-making.

CRediT authorship contribution statement

Eric Augusto Melchor Cruz: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Luis Alberto Hernández Armenta:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation. **Mohamed Badaoui:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors gratefully acknowledge the National Center of Energy Control (CENACE) for its support in the validation of the finding of this research.

Data availability

Data will be made available on request.

References

- [1] W. Morais, R. Perez, A. Soares, Aplicación de un modelo de expansión de la generación considerando el requerimiento de reserva probabilística dinámica en el sistema eléctrico mexicano, in: Foz do Iguaçu, Brasil: XVIII ERIAC DÉCIMO OITAVO ENCONTRO REGIONAL IBERO-AMERICANO DO CIGRE, 2019 [Online]. Available: <https://www.psr-inc.com/es/aplicacion-de-un-modelo-de-expansion-de-la-generacion-considerando-el-requerimiento-de-reserva-probabilistica-dinamica-en-el-sistema-electrico-mexicano/>.
- [2] P. McSharry, S. Bouwman, G. Bloemhof, Probabilistic forecasts of the magnitude and timing of peak electricity demand, *IEEE Trans. Power Syst.* 20 (2) (2005) 1166–1172, <https://doi.org/10.1109/TPWRS.2005.846071>.
- [3] R.J. Hyndman, S. Fan, Density forecasting for long-term peak electricity demand, *IEEE Trans. Power Syst.* 25 (2) (2010) 1142–1153, <https://doi.org/10.1109/TPWRS.2009.2306017>.
- [4] Z. Wang, F. Kong, S. Feng, M. Wang, X. Yang, H. Zhao, D. Wang, Y. Zhang, Is mamba effective for time series forecasting? [Online]. Available: <https://arxiv.org/abs/2403.11144>, 2024.
- [5] K. Lindberg, P. Seljom, H. Madsen, D. Fischer, M. Korpås, Long-term electricity load forecasting: current and future trends, *Util. Policy* 58 (2019) 102–119, <https://doi.org/10.1016/j.jup.2019.04.001> [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957178719300116>.
- [6] G. Memarzadeh, F. Amirteimoury, H. Noori, F. Keynia, An electrical load forecasting model based on a novel closed loop neural networks and interaction gain feature selection, *Results Eng.* 27 (2025) 105800 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2590123025018717>.
- [7] C. Wang, J. Song, D. Shi, J.L. Reyna, H. Horsey, S. Feron, Y. Zhou, Z. Ouyang, Y. Li, R.B. Jackson, Impacts of climate change, population growth, and power sector decarbonization on urban building energy use, *Nat. Commun.* 14 (1) (2023) 1–16, <https://doi.org/10.1038/s41467-023-41458-5> [Online].
- [8] A. Werlang, G. Cunha, J. Bastos, J. Serra, B. Barbosa, L. Barroso, Reliability metrics for generation planning and the role of regulation in the energy transition: case studies of Brazil and Mexico, *Energies* 14 (21) (2021) [Online]. Available: <https://www.mdpi.com/1996-1073/14/21/7428>.
- [9] G. Ladan, M. Kalantar, Different methods of longterm electric, load demand forecasting; a comprehensive review, *Iran. J. Electr. Electron. Eng.* 7 (4) (2011) 249–259.
- [10] C. Yin, K. Liu, Q. Zhang, K. Hu, Z. Yang, L. Yang, N. Zhao, SARIMA-based medium- and long-term load forecasting, *Strateg. Plan. Energy Environ.* (2023), <https://doi.org/10.13052/spee1048-5236.4222>.
- [11] Y. Yang, J. Guo, Y. Li, J. Zhou, Forecasting day-ahead electricity prices with spatial dependence, *Int. J. Forecast.* 40 (3) (2024) 1255–1270, <https://doi.org/10.1016/j.ijforecast.2023.11.006> [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0169207023001152>.
- [12] H. Al-Shaikh, M.A. Rahman, A. Zubair, Short-term electric demand forecasting for power systems using similar months approach based SARIMA, in: 2019 IEEE International Conference on Power, Electrical, and Electronics and Industrial Applications (PEEIACON), Nov. 2019, pp. 122–126.
- [13] M. Zhang, M.-A. Millar, S. Chen, Y. Ren, Z. Yu, J. Yu, Enhancing hourly heat demand prediction through artificial neural networks: a national level case study, *Energy AI* 15 (2024) 100315 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666546823000873>.
- [14] R. Gupta, A.K. Yadav, S. Jha, P.K. Pathak, Predicting global horizontal irradiance of north central region of India via machine learning regressor algorithms, *Eng. Appl. Artif. Intell.* 133 (2024) 108426 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0952197624005840>.
- [15] N. Shahsavari-Pour, A. Heydari, F. Keynia, A. Fekih, A. Shahsavari-Pour, Building electrical consumption patterns forecasting based on a novel hybrid deep learning model, *Results Eng.* 26 (2025) 104522 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2590123025005997>.
- [16] L. Tighitiz, J. Yoo, W.K. Al-Shibli, Strategic enhancements in electricity price forecasting: the role of xgboost and error correction features, *Results Eng.* 26 (2025) 105609 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2590123025016792>.
- [17] S. A. M.S. Christo, J.V. Elizabeth, A hybrid approach to time series forecasting: integrating arima and prophet for improved accuracy, *Results Eng.* 27 (2025) 105703 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2590123025017748>.
- [18] R. Gupta, A.K. Yadav, S. Jha, P.K. Pathak, Composition of feature selection techniques for improving the global horizontal irradiance estimation via machine learning models, *Therm. Sci. Eng. Prog.* 48 (2024) 102394 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S245190492400012X>.
- [19] A. Aksöz, S. Oyucu, E. Biçer, R. Bayındır, Analysis of SARIMA models for forecasting electricity demand, in: 2024 12th International Conference on Smart Grid (icSmart-Grid), 2024, pp. 767–771.
- [20] J.A. Zancanaro, O.L. dos Santos, L.F. Ugarte, M. Giesbrecht, M.C. de Almeida, Energy consumption forecasting using sarima and narnet: an actual case study at university campus, in: 2019 IEEE PES Innovative Smart Grid Technologies Conference - Latin America (ISGT Latin America), 2019, pp. 1–6.
- [21] QiLyu QiheLou, Z. Na, D. Ma, X. Ma, Short-term electric power demand forecasting using a hybrid model of SARIMA and SVR, *IOP Conf. Ser. Earth Environ. Sci.* 619 (1) (Dec. 2020) 012035, <https://doi.org/10.1088/1755-1315/619/1/012035> [Online]. Available: <https://doi.org/10.1088/1755-1315/619/1/012035>.
- [22] R. Gupta, A.K. Yadav, S. Jha, P.K. Pathak, Long term estimation of global horizontal irradiance using machine learning algorithms, *Optik* 283 (2023) 170873 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0030402623003698>.
- [23] A. Kumar Dubey, A. Kumar, V. García-Díaz, A. Kumar Sharma, K. Kanhaiya, Study and analysis of SARIMA and LSTM in forecasting time series data, *Sust. Energy Technol. Assess.* 47 (2021) 101474, <https://doi.org/10.1016/j.seta.2021.101474> [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2213138821004847>.
- [24] A.V. Mendoza, J.H. Tovar Hernández, H.F. Ruiz Paredes, Electricity demand forecasting for the BCS region of the Mexican SEN using artificial neural networks, in: 2023 IEEE International Autumn Meeting on Power, Electronics and Computing (ROPEC), vol. 7, 2023, pp. 1–6.
- [25] J. Arenas-López, J. Martínez-Martínez, M. Badaoui, R. Mota-Palomino, Modelos de demanda de electricidad en tiempo continuo y en tiempo discreto, *Rev. Int. Invest. Innov. Tecnol.* (ISSN 2007-9753) (2022).
- [26] M.A. Luna, G.G. Tobón, R. Arenas, M. Cortés, R. Ciprian, F. Ortiz, The relevance and importance of the demand and consumption forecast in the long-term planning of electrical system, *CIGRE* (2024), c1-11046-2024.
- [27] J.P. Arenas-López, M. Badaoui, The Ornstein-Uhlenbeck process for estimating wind power under a memoryless transformation, *Energy* 213 (2020) 118842, <https://doi.org/10.1016/j.energy.2020.118842> [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0360544220319496>.
- [28] Operación del sistema eléctrico nacional [Online]. Available: <https://www.cenace.gob.mx/Paginas/SIM/OperacionSEN.aspx>. (Accessed 1 September 2024).
- [29] R. Lobato-Sánchez, El monitor de la sequía en México, *Tecnol. Cienc. Agua* 7 (5) (2016) 197–211.
- [30] P.J. Brockwell, R.A. Davis, *Introduction to Time Series and Forecasting*, Springer, 2002.
- [31] C. Chatfield, *The Analysis of Time Series: An Introduction*, sixth edition, Chapman & Hall/CRC Texts in Statistical Science., CRC Press, 2016.
- [32] A. Leon-García, *Probability, Statistics, and Random Processes for Electrical Engineering*, Pearson Education, 2017.
- [33] Z. Leonowicz, T. Lobos, P. Schegner, Modern spectral analysis of non-stationary signals in electrical power systems, in: 14th Power System Computation Conference, Sevilla, Spain, 2002.
- [34] C. Shakila, S. Khadar Babu, Evaluating and contrasting machine learning and statistical techniques for time series forecasting with hyperparameter optimization, *Results Eng.* 27 (2025) 106275 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2590123025023473>.
- [35] J. Banaś, K. Utnik-Banaś, Evaluating a seasonal autoregressive moving average model with an exogenous variable for short-term timber price forecasting, *For. Policy Econ.* 131 (2021) 102564, <https://doi.org/10.1016/j.forpol.2021.102564>.
- [36] J.J.M. Moreno, A.P. Pol, A.S. Abad, B.C. Blasco, Using the r-mape index as a resistant measure of forecast accuracy, *Psicothema* 25 (4) (2013) 500–506.